

FANUC Robot **series**

R-30iB, R-30iB Plus **CONTROLLER** Version 9.10 and later

ZDT **OPERATOR'S MANUAL**

MARUGZDTA10191E REV B

ORIGINAL INSTRUCTIONS

Thank you very much for purchasing a FANUC robot.

Before using the robot, be sure to read the, *FANUC Robot SAFETY HANDBOOK (B-80687EN)* and understand its contents.

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In this manual, we endeavor to include all pertinent matters. There are, however, a very large number of operations that must not or cannot be performed. Please assume that any operations that are not explicitly described as being possible are not possible.

SAFETY

Safety Precautions

This chapter must be read before using the robot. For detailed functions of the robot operation, read the relevant operator's manual to understand fully its specification.

For the safety of the operator and the system, follow all safety precautions when operating a robot and its peripheral equipment installed in a work cell. For safe use of FANUC robots, you must read and follow the instructions in the *FANUC Robot SAFETY HANDBOOK (B-80687EN)*.

Definition of User

Personnel can be classified as follows:

Operator • Turns the robot controller power on/off • Starts the robot program from operator panel
Programming or Teaching Operator • Operates the robot • Teaches the robot inside the safeguarded space Maintenance engineer
Maintenance Technician • Operates the robot • Teaches the robot inside the safeguarded space • Maintenance (repair, adjustment, replacement)

The operator is not allowed to work in the safeguarded space.

The programmer or teaching operator and maintenance engineer are allowed to work in the safeguarded space. Work carried out in the safeguarded space include transportation, installation, teaching, adjustment, and maintenance.

To work inside the safeguarded space, the person must be trained on proper robot operation.

Table s-1 lists the work outside the safeguarded space. In this table, the symbol “○” means the work is allowed to be carried out by the worker.

Table s-1 Work Performed Outside the Safeguarded Space

	Operator	Programmer or Teaching Operator	Maintenance Technician
Turn power ON/OFF to Robot controller	○	○	○
Select operating mode (AUTO, T1, T2)		○	○
Select remote/local mode		○	○
Select robot program with teach pendant		○	○

Select robot program with external device		○	○
Start robot program with operator's panel	○	○	○
Start robot program with teach pendant		○	○
Reset alarm with operator's panel		○	○
Reset alarm with teach pendant			
Set data on teach pendant			
Teaching with teach pendant			
Emergency stop with operator's panel	○	○	○
Emergency stop with teach pendant	○	○	○
Operator Panel maintenance		○	
Teach Pendant maintenance			○

During robot operation, programming and maintenance, the operator, programmer, teaching operator and maintenance engineer take care of their own safety using at least the following safety protectors:

- Use clothes, uniform, overall adequate for the work
- Safety shoes
- Helmet

Definition of Safety Notations

To ensure the safety of workers and prevent damage to the machine, this manual indicates each precaution on safety according to its severity identified below. Read the contents of each **Warning** and **Caution** before attempting to use the robot.

 WARNING
Indicates a hazard resulting in the death or serious injury of the user could occur if he or she fails to follow the approved procedure.
 CAUTION
Indicates a hazard resulting in the injury of the user or damage to equipment could occur if the user fails to follow the approved procedure.

Note

Indicates a supplementary explanation not related to any WARNING or CAUTION.

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PART I

ROBOT SETUP

Topics:

- [ZDT OVERVIEW](#)
- [ZDT COMMUNICATION AND BASIC SETUP](#)
- [ZDT Health Functions and Their Setup](#)
- [Internal Errors](#)

This section describes how to setup ZDT, particularly with regard to the robot.

1 ZDT OVERVIEW

Customer's facilities run at high capacity, often twenty-four hours a day, seven days a week with little or no time allocated for proper maintenance or to inspect the equipment. This can lead to unexpected downtime. Any unexpected downtime can become very expensive as it represents the potential loss of production.

The goal of Zero Down Time (ZDT) is to prevent unexpected downtime related to a robot or robot system malfunction. Key features include:

- The ability to predict potential equipment failures associated with the robot, controller, and processes controlled by the robot.
- Advanced analytics to facilitate equipment optimization.
- Smart maintenance notifications to extend equipment life and to help minimize maintenance costs.

ZDT features cover the following areas:

- Mechanical health
Includes: Robot drivetrain, sudden stop events, etc.
- Process health
Includes: Process specific items.
- System health
Includes: Alarm information, memory usage, program and variable changes, etc.
- Maintenance Health
Includes intervals for robot, grease change, battery exchange, etc.

1.1 ZDT components

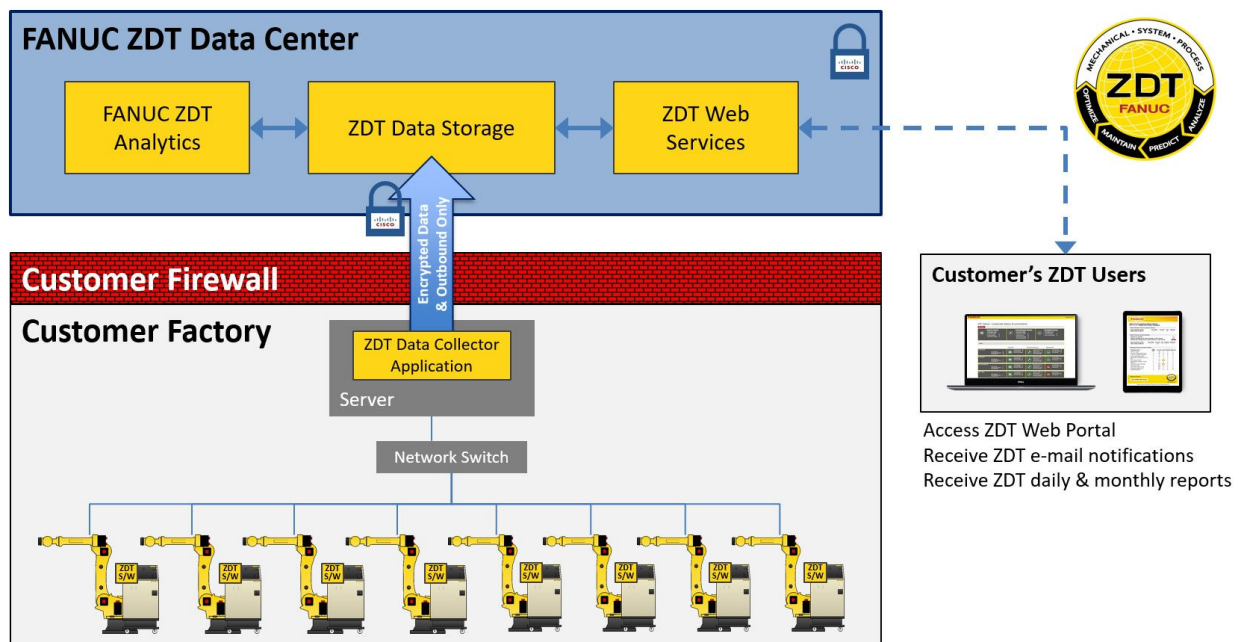


Figure 1.1 ZDT Overview

1.1.1 Robot Software

On the robot controller, ZDT collects specific data generated during normal production or generated by periodically executed self-test programs. The self-test programs typically take less than two minutes to run and are designed to be executed during non-production time. The data collected is minimal and with focus on the ZDT health issue being monitored.

ZDT is designed to have ZERO impact on robot execution and minimal effect on memory storage.

The data on the robot controller is embedded into messages that are sent to a FANUC Data Collector. In some cases, such as paint systems using PaintWorks, the data is sent or gathered by a PC and then placed into messages to be sent to the Data Collector.

1.1.2 ZDT Data Collector

ZDT Data Collectors receive data from networked FANUC robots, then packages and sends the data securely to the FANUC Data Center with encryption. An outbound firewall port from Data Collector is required for secure transmission to the ZDT Data Center. ZDT data flow has low network bandwidth requirements.

1.1.3 FANUC Data Center

At the FANUC Data Center, analytics are performed based on the design of FANUC engineers. If an analytic detects a potential issue, an alert is issued and the customer is notified. This alert will indicate the specific condition detected, associated data, and the action to be taken. The goal of ZDT is to provide several days of advanced warning so the issue can be addressed during a scheduled stoppage, allowing production to continue until the next scheduled downtime.

Use of the cloud is advantageous in that it allows altering of analytics to improve prediction capabilities without causing any disruption or risk to production.

1.1.4 Web Portal

A web portal is available for customers to view charts of data and analytic results by plant, production zone, or robot at any time.

2 ZDT COMMUNICATION AND BASIC SETUP

The robot controller provides a setup screen to enable/disable the sending of ZDT data to the data collector for selected ZDT Health suites. As more ZDT Health features are added, this screen will expand so the actual menu may differ from what is shown in this document.

2.1 ZDT Client Setup on the TP

To access the Setup ZDT Client screen perform the following: **MENU > SETUP > ZDT Client**.

A screen similar to [Figure 2.1](#) will display.

The screenshot shows the ZDT Client Setup screen with a top status bar and a bottom navigation bar. The status bar includes buttons for 'Busy', 'Step', 'Hold', 'Fault', 'Run', 'I/O', 'Prod', and 'TCyc', along with a message 'MOTN-170 Load is close to capacity', a 'T2' indicator, 'JOINT' text, and a '100%' battery level. The main area is split into two panes. The left pane, titled 'SETUP ZDT Client', lists 16 items with their status: 1 ZDT Client: ENABLED, 2 Use XML Config: ENABLED, 3 Periodic Status: <*<DETAIL*>, 4 Alarms: ENABLED, 5 TP Change Log: DISABLED, 6 Var Change Log: DISABLED, 7 Process Logger Monitor: Not Loaded, 8 Vision Images: Not Loaded, 9 iRDiagnosics: <*<DETAIL*>, 10 System Clock: <*<DETAIL*>, 11 Application Status: ENABLED, 12 Maintenance Reminder: <*<DETAIL*>, 13 iRPickTool: Not Loaded, 14 Arc Weld Health: Not Loaded, 15 Servogun Health: Not Loaded, and 16 HTTP Config: <*<DETAIL*>. The right pane, titled 'SETUP http', shows configuration details: HTTP: ENABLED, User: test, Password: *****, URL: http://172.22.194.39:6063/zdtdata/, UID: c962b37a-lac0-eb2a-f1c7-8c6eb548c816, 1 Last Err: 0, Last Sent: 25-OCT-19 10:36:14 AM, POST Stat: GINFO_P [SENT], 2 Retry Connection: DISABLED, and Ethernet Port: Port #1. At the bottom of the right pane, it says 'Fields initialized by FR:\RCMCFG.XML'. The bottom navigation bar contains a grid icon, '[TYPE]', 'DISP_CFG', 'DISP_VAR', 'DISP_ERR', and two empty buttons.

Item	Status
1 ZDT Client:	ENABLED
2 Use XML Config:	ENABLED
3 Periodic Status:	<*<DETAIL*>
4 Alarms:	ENABLED
5 TP Change Log:	DISABLED
6 Var Change Log:	DISABLED
7 Process Logger Monitor	Not Loaded
8 Vision Images:	Not Loaded
9 iRDiagnosics:	<*<DETAIL*>
10 System Clock:	<*<DETAIL*>
11 Application Status:	ENABLED
12 Maintenance Reminder:	<*<DETAIL*>
13 iRPickTool:	Not Loaded
14 Arc Weld Health:	Not Loaded
15 Servogun Health:	Not Loaded
16 HTTP Config:	<*<DETAIL*>

HTTP: ENABLED
User: test
Password: *****
URL: http://172.22.194.39:6063/zdtdata/
UID: c962b37a-lac0-eb2a-f1c7-8c6eb548c816
1 Last Err: 0
Last Sent: 25-OCT-19 10:36:14 AM
POST Stat: GINFO_P [SENT]
2 Retry Connection: DISABLED
Ethernet Port: Port #1

Fields initialized by FR:\RCMCFG.XML

Figure 2.1 Setup ZDT Client screen

2.1.1 Enable Disable ZDT

To enable ZDT, cursor to **ZDT Client:** and choose **ENABLE**. If for some reason (e.g. decommissioning a Data Collector) you wish to disable ZDT, cursor to **ZDT Client:** and choose **DISABLE**.

2.1.2 Enable ZDT Features

Press **i + FCTN** to display details of each selection.

Unless informed by FANUC, please **ENABLE** all ZDT features. Some features, when enabled in the left pane, have associated items in the right pane that also need to be enabled.

Note

Please make sure the system clock is set properly.

Several parameters are used to configure ZDT. [Table 2.1.2](#) briefly describes all the parameters.

Table 2.1.2 ZDT Screen Items

Item	Description
ZDT Client	Enable or Disable ZDT. Items in the detail page are also set by the <code>RCMCFG.XML</code> file if USE XML CONFIG is enabled. Remember to move the cursor to HTTP Config to setup the http configuration.
Use XML Config	This item, when enabled, will force ZDT to use the configuration information found within the <code>RCMCFG.XML</code> file. Using an XML file is typically used in installations with large number of robot. Once enabled, the configuration items become inactive and only can be modified by XML. Once it is enabled, all configuration items are populated from <code>RCMCFG.XML</code> immediately. To manually configure ZDT this item must be disabled.
Periodic Status	Pressing <*DETAIL*> will show in the left screen when basic controller information will periodically be sent to the data collector. The default is 86400 seconds which is 24 hours. This number should not be changed unless instructed by a FANUC engineer.
Alarms	This item, if ENABLED, will permit alarms to be sent to the ZDT data center. If NONE of the categories are enabled, all alarms with PAUSE, ABORT, STOP and SERVO severity will be sent.
TP Change Log	This item, if ENABLED, will send TP program changes to the data center.
Var Change Log	This item, if ENABLED, will send configured variable (<code>VARCHG.XML</code>) changes to the data center. Configuration details are available in Section 3.2, System Health .
Process Logger Monitor	This item brings up the related view to configure Process Logger Monitor for ZDT. If the option is not loaded, then it shows <code>Not Loaded</code> . Please review the Process Logger manual for more information concerning process logger.

Item	Description	
Vision Images	This item brings up the related view to configure Vision setup for ZDT. Vision data will automatically be sent at runtime if enabled. In addition, you may control sending ZDT data by calling the IRVIZDT macro from the TP program. If <i>iR</i> Vision is not loaded, then it shows <code>Not Loaded</code> .	
	Item	Explanation
	Enable ZDT for <i>iR</i> Vision	When this item is checked, the system will automatically queue vision ZDT messages whenever any vision runtime data is generated.
	Disable Images for ZDT	When this item is checked, the system does not perform any processing related to saving images for ZDT function.
	ZDT Image Buffer Size	The maximum size of images to be sent to the ZDT data collector. [640×512] When the camera image is larger than this size, the image will be shrunk to be smaller than this size, and then sent to the ZDT data collector. It will reduce the network traffic and the capacity load of the ZDT server. [1280×1024] By default, this size is selected.
	ZDT send period (seconds)	This item configures the period at which vision ZDT messages are sent to the server. The range is 60 to 600 seconds, and the default value is 60.
	Message byte limit	This item configures the number of bytes which will be sent from the vision ZDT queue per period. It is used in order to limit the amount of data being sent to ZDT, thus reducing any performance impact it may have on the system. The range is 50000 to 10000000, and the default value is 50000.
	Message queue size	This item configures the number of vision data results the queue holds. The range is 10 to 70, and the default value is 10.
	Select Vision Process	Select the vision process whose threshold values should be reset.
	Select Analytic	Select the vision analytic which should be reset.
	Select Threshold	Select the threshold type which should be reset.
	Use Existing Data for calculation	Check this box if you wish to use the existing ZDT server data to re-calculate the threshold. If unchecked, the ZDT server will use only future results to calculate the new threshold.

Item	Description		
iRDiagnostics	This item brings up the related view to configure ZDT iRDiagnostics.		
	Item	Explanation	
	Robot Condition	1 Event Data	Send data after DIAG (Self-test) program executes
		2 Daily Data	Send data daily
	Servo Off Events	3 Servo Off Events	Enable/disable sending servo off events
	High Water Mark Job data	4 Job Data	Enable/Disable sending Job Data to ZDT Data Center
		5 Mode	Triggering Mode. Refer to Section 3.1.3, Job Data (High Water Mark Job Data) .
System Clock	This item brings up the related view to configure the System Clock. It may seem basic, however it is important for the correct time to be sent with each message to correlate events on the FAC Data center. Verify the System Clock is set correctly and use the ADV function key to verify the Daylight Saving Time and Time zone settings.		
Application Status	This item brings up the related view to configure the periodic Application data for ZDT Status.		
	Item	Explanation	
	In-Cycle Measured	Choose Busy or Motion	
	Cycle Count by	Four Choices are available: 1. <i>Program Run</i> : Count the number of Programs Runs 2. <i>UOP Prod_Start</i> : Count the number of Prod_Start signals from UOP 3. <i>Input Port</i> : Count the number of OFF/On of specified DI 4. <i>User Program</i> , use specified register	
	Register Index	Register number if you select User Program above.	
	Current Cycle Count	Current Count accumulated	
Maintenance Reminder	This item brings up the Maintenance Reminder configuration screen, if the option is loaded. If the option is not loaded, then it shows <code>Not Loaded</code> .		
	Item	Explanation	
	Send ZDT Message	If ENABLED, send maintenance reminder data	
	Send ZDT message Alert	If ENABLED, send data when maintenance reminder alarm occurs	
	Periodic Timer	Time interval (hours) of which data is sent to the ZDT Data Center	
iRPickTool	This item brings up the iRPickTool configuration screen if the option is loaded. If the option is not loaded, then it shows <code>Not Loaded</code> .		

2.2 ZDT Configuration

2.2.1 XML Configuration File

To set up ZDT using a custom configuration file, modify `RCMCFG.XML` and load it using the file menu or load it using the load function. See [Section 2.2.4, Loading RCMCFG.XML Configuration File](#). The custom file is used instead of the default file after the file is loaded. Once the custom file is loaded, then **FCTN > SAVE** copies `RCMCFG.XML` to the default device. This file is also backed up and restored.

ZDT has a default `RCMCFG.XML` file. The file is copied to a default device when the **FCTN > SAVE** operation is selected while on the ZDT menu.

To disable the configuration of ZDT by `RCMCFG.XML`, disable the capability in the ZDT client menu. This will allow for all configuration items to be configurable manually. Please note that the configuration file still exists, but it is not in use. So you can enable and reuse any time later without reloading the file.

2.2.2 RCMCFG.XML Tag Definitions

The tag definitions for the `RCMCFG.XML` file are described in [Table 2.2.2](#).

Table 2.2.2 Tag Definitions

Tag	Parent Tag	Depth	Definition
<RCMCFG>	None	0	This is root tag.
<PLANT>	<RCMCFG>	1	Tag to specify the plant information. It has one attribute: <i>name</i> unique across plants
<LINE>	<PLANT>	2	Tag to specify the line information in the plant. It is child of <PLANT> tag. It has one attribute: <i>name</i> attribute to specify name of the lines in the plant.
<CLUSTER>	<LINE>	3	Tag to specify cluster information of the line in the plant. It is child of <LINE> tag. It has three attributes: <i>name</i> attribute to specify name of the cluster. <i>ipgroup</i> (required) attribute to specify the first three octets of IP address. <i>ipset</i> (required) defines the SET of last octet of IP addresses. The <i>ipset</i> could be specified in two ways, either in range if all robots in cluster has continuous IP addresses e.g. <code>ipset="200-300"</code> or in set of last octets otherwise e.g. <code>ipset="123,178, 200, 201, 223"</code>. A controller will be assigned the first line and cluster that matches the controller's IP address.
<ETHPORT>	<RCMCFG>	1	Tag to specify the Ethernet port to be used. Port1 or Port2 is expected.

Tag	Parent Tag	Depth	Definition
<HTTP>	<RCMCFG>	1	This tag defines FANUC ZDT Data Collector details. 1. <i>disable</i> attribute enables or disables HTTP client. If it is set to 0 then HTTP will be used to route the data and E-mail is used if it is 1. If this is not available, then it is enabled by default. So it is only needed if you need to disable the HTTP.
<URL>	<HTTP>	2	Tag to define the FANUC ZDT Data Collector URL. It has the attribute <i>method</i> which should have value "POST".
<AUTH>	<HTTP>	2	<i>Normally not used.</i> This defines the FANUC ZDT Data Collector authentication details. 1. <i>user</i> attribute defines user name. 2. <i>pwd</i> attribute defines FANUC ZDT Data Collector password.
<SEVERITY>	<RCMCFG>	1	Defines ALL errors of this severity. It has four attributes: 1. <i>name</i> of severity - "NONE", "WARN", "PAUSE", "STOP", "SERVO", "SERVO2", "ABORT", "SYSTEM", "NO DISP", " 2. <i>global</i> - If TRUE then all errors have global severity. If FALSE then all errors have local severity. This attribute applies only to "PAUSE", "STOP", and "ABORT" severities, otherwise the attribute is ignored. 3. <i>send</i> - If TRUE then E-mail is sent immediately on that error. 4. <i>alert</i> - If TRUE then E-mail is sent immediately on that error and alert text message is sent to phone.
<CATEGORY>	<RCMCFG>	1	<i>Normally not used.</i> Defines ALL errors of this category. It has three attributes: 1. <i>name</i> of category – "MOTION", "APPL", "PWD", "\COMM", "SYSTEM" 2. <i>send</i> - If TRUE then E-mail is sent immediately on that error. 3. <i>alert</i> -If TRUE then E-mail is sent immediately on that error and alert text message is sent to phone.
<FILE>	<SEVERITY> <CATEGORY> <ERROR>	2/3	<i>Normally not used.</i> Defines files to be attached with error. File name is included in tag text not in attribute.
<EMBED>	<SEVERITY> <CATEGORY> <ERROR>	2/3	<i>Normally not used.</i> Defines files to be embedded in E-mail body with specified error. File name is included in tag text not in attribute.
<ERTYPE>	<RCMCFG>	1	<i>Normally not used.</i> Defines specific Error category i.e. facility code. It has <i>name</i> attribute to specify facility code.

Tag	Parent Tag	Depth	Definition
<ERROR>	<ERTYPE>	2	<i>Normally not used.</i> Defines specific errors. It has seven attributes: 1. <i>code</i> to specify error number. It could be '*' if <i>sev</i> is specified. 2. <i>sev</i> to define error severity. 3. <i>global</i>-If TRUE then all errors which have global severity. If FALSE then all errors with local severity. 4. <i>send</i>- If TRUE then E-mail is sent immediately on that error. 5. <i>alert</i>-If TRUE then E-mail is sent immediately on that error and alert text message is sent to phone. 6. <i>freq</i> to define frequency of an alarm before sending. 7. <i>dur</i> to define duration of time before sending an alarm.

2.2.3 Sample configuration file

Below is a sample RCMCFG.XML file. The filename must be RCMCFG.XML and must reside in FR:

```
<!-- This file is processed one time at startup. Tags are always
English. ->
<?xml version="1.0" ?>
<RCMCFG>
  <!-- Please edit this info as per your plant -->
  <PLANT name="plant">
    <LINE name="line">
      <CLUSTER name="cluster" ipgroup="172.30.1" ipset="0-254" />
    </LINE>
  </PLANT>
  <ETHPORT> Port2</ETHPORT>
  <HTTP disable="0">
    <URL>method="POST">http://172.30.1.40:6063/ZDTData/</URL>
    <AUTH user="test" pwd="password"/>
  </HTTP>
  <!-- Power message, one of its availability invokes periodic
status message ->
  <ERTYPE name="SYST">
    <ERROR code="26">
      <FILE name="MD:errhist.ls"/>
      <FILE name="MD:errmot.ls"/>
    </ERROR>
  </ERTYPE>
  <!-- Severity based configuration ->
  <SEVERITY name="PAUSE" send="TRUE"/>
  <SEVERITY name="STOP" send="TRUE"/>
  <SEVERITY name="ABORT" send="TRUE"/>
  <SEVERITY name="SERVO" send="TRUE"/>
  <!-- Messaging library settings -->
  <!-- *WARNING* Make sure maxinst < 128!! it gets converted to
signed byte then ushort -->
  <MSGLIB maxmem="204800" msglim="10240">
    <MSG varname="GCOND_P" initinst="5" maxinst="5" prior="1"/>
    <MSG varname="GSRVOF_BIN_P" initinst="5" maxinst="5" prior="1"/>
    <MSG varname="GSRVOF_DTL_P" initinst="5" maxinst="5" prior="1"/>
  </MSGLIB>
</RCMCFG>
```

```
<MSG varname="GJOB_DATA_P" initinst="5" maxinst="5" prior="1"/>
<MSG varname="GFLEX_DATA_P" initinst="5" maxinst="5" prior="1"/>
<MSG varname="GFLEX_IO_P" initinst="5" maxinst="5" prior="1"/>
<MSG varname="GODOMETER_P" initinst="5" maxinst="5" prior="1"/>
<MSG varname="CINFO_P" initinst="1" maxinst="5" prior="0"/>
<MSG varname="GINFO_P" initinst="1" maxinst="8" prior="0"/>
<MSG varname="VARCHG_P" initinst="1" maxinst="10" prior="0"/>
<MSG varname="ALARM_MSG_P" initinst="1" maxinst="20" prior="0"/>
<MSG varname="TPPRG_CHG_P" initinst="1" maxinst="10" prior="0"/>
<MSG varname="CMEMORY_P" initinst="1" maxinst="5" prior="0"/>
<MSG varname="RTC_P" initinst="1" maxinst="5" prior="0"/>
<MSG varname="MBE_P" initinst="1" maxinst="5" prior="0"/>
<MSG varname="SVGUN_DIAG_P" initinst="5" maxinst="5" prior="0"/>
<MSG varname="SVGUN_CAP_P" initinst="5" maxinst="5" prior="0"/>
<MSG varname="PLOG_STAT_P" initinst="1" maxinst="50" prior="0"/>
<MSG varname="APP_DATA_P" initinst="5" maxinst="5" prior="0"/>
<MSG varname="GMAINT_ITM_P" initinst="1" maxinst="50" prior="0"/>
<MSG varname="GMAINT_EVT_P" initinst="1" maxinst="50" prior="0"/>
<MSG varname="GMNT_HEAD_P" initinst="1" maxinst="5" prior="0"/>
<MSG varname="VIS_RUNDAT_P" initinst="1" maxinst="10" prior="1"/>
<MSG varname="VIS_RUNRES_P" initinst="1" maxinst="100"
prior="1"/>
<MSG varname="IRC_GNRC_P" initinst="1" maxinst="50" prior="1"/>
<MSG varname="BATT_LOW2_T" initinst="1" maxinst="5" prior="0"/>
</MSGLIB>
</RCMCFG>
```

 CAUTION

Never copy the configuration file from this document because sometimes hidden characters sneak in and it causes failure in XML file parsing. Always pull the default file from the controller and modify it. You can do MD: backup and get it or directly save it using the SAVE AS function.

2.2.4 Loading RCMCFG.XML Configuration File

This section shows two ways to load the Configuration file.

Loading from the File Menu

You can load RCMCFG.XML directly from the file menu. The file name **MUST** be RCMCFG. Press **MENU > FILE** on the teach pendant, select the file, and then select **LOAD**.

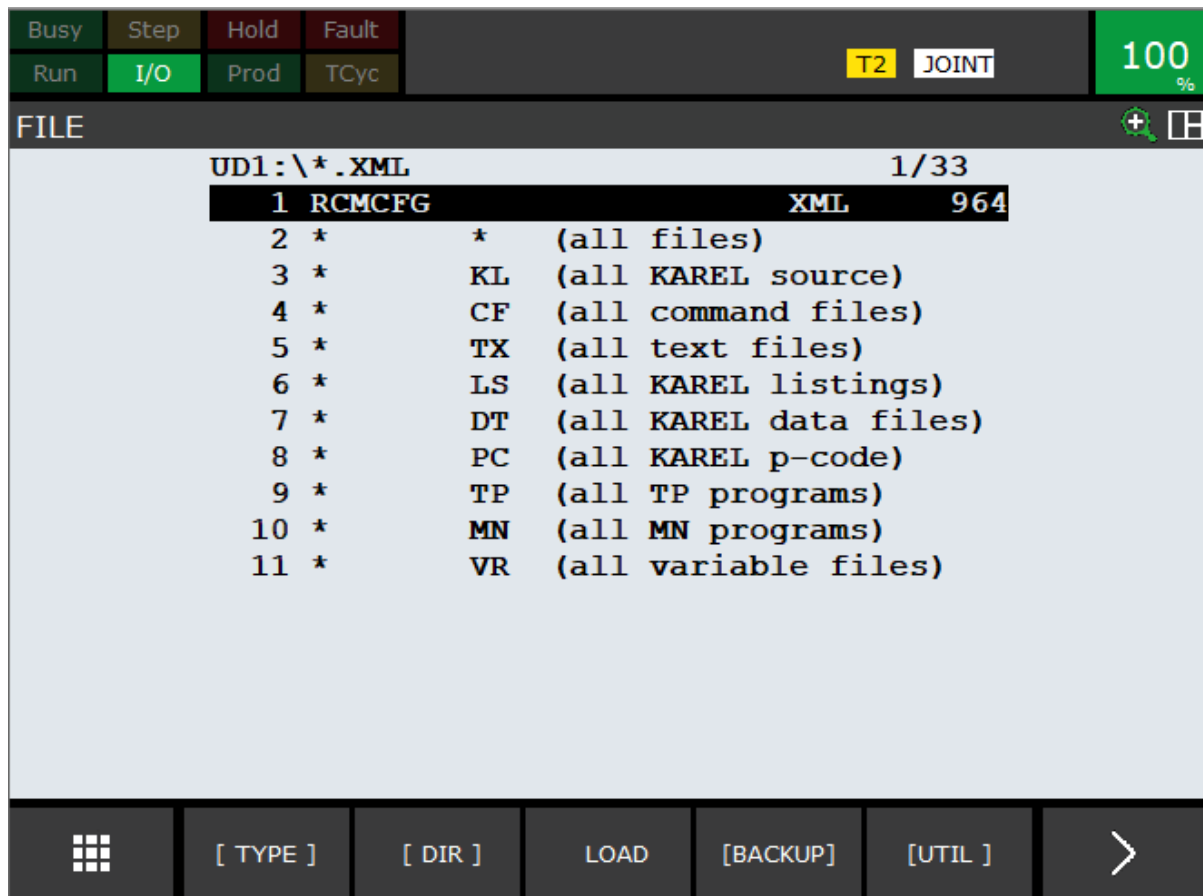


Figure 2.2.4 File Menu

Loading from ZDT Client Menu

Another alternative to load the file is from the ZDT screen. This procedure allows you to use the file name of your choice. Perform the steps in the following procedure.

2.2.4.1 Importing rcmcfg.xml File

About this task

Use this procedure to import the `rcmcfg.xml` file.

Procedure

1. Press the **DISP** key to switch focus to the **SETUP ZDT Client** pane (Left pane).
2. Select **HTTP Config**, press **ENTER** to enter the HTTP detail config screen.
3. Press the **DISP** key to switch focus to the **SETUP http** pane (Right pane).
4. Select **F2, IMPORT**.
5. Choose **F4, [RCMCFG]**. You should see a screen similar to [Figure 2.2.4.1](#).

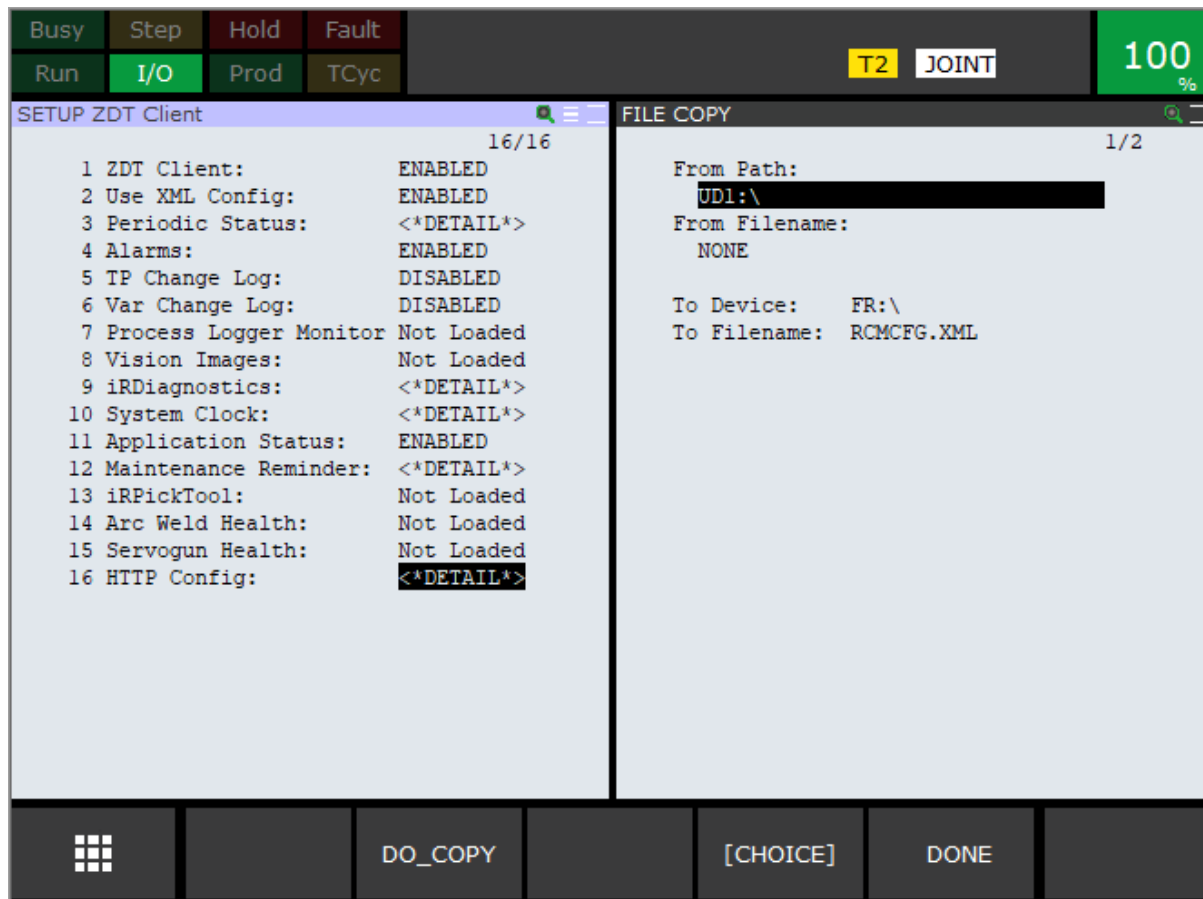


Figure 2.2.4.1 Import File

6. Select **From Path**, press **F4, CHOICE**.
7. Choose the Device where `rcmcfg.xml` is stored.
8. Select **From Filename**, press **F4, CHOICE**.
9. Choose **RCMCFG.XML**.
10. Select **F2, DO_COPY**. If prompted to overwrite, press **F4, YES**.
11. Screen will show `Copied to FR:\RCMCFG.XML`.
12. Select **F5, DONE**.
13. Select **F3, REININT**. Screen will show `REINIT Successful`.

Note

When REINIT is successful, CINFO and GINFO messages are sent to the cloud to provision the robot in the ZDT portal. **REINIT** can be pressed at any time to send CINFO and GINFO messages to the cloud, so the user does not have to wait for 24 hours for the messages to be sent.

2.3 Validating the Installation of the rcmcfg.xml file

About this task

This task allows you to verify the Server IP Address Port and End Point.

Procedure

1. Press **MENU > SETUP > ZDT Client** to go to **SETUP ZDT Client** Screen.
See previous steps in [Section 2.1, ZDT Client Setup on the TP](#) for help to enter the related views with the ZDT Client setup.
2. Select **HTTP Config**, press **ENTER**.
3. Validate that Data Collector IP Address matches the plant's data collector IP address.
4. The server can be validated using the **F1, TEST** button on the **SETUP http** screen.

Note

At the end of the installation you might see the message `Configuration Error (FR:\RCMCFG.XML)`. This can be ignored.

Note

When TEST is successful, CINFO and GINFO messages are sent to the cloud to provision the robot in the ZDT portal. **TEST** can be pressed at any time to send CINFO and GINFO messages to the cloud, so the user does not have to wait 24 hours for the messages to be sent.

3 ZDT Health Functions and Their Setup

This chapter describes ZDT features and the necessary steps that need to be performed to set up and use them.

Note

New ZDT features are added often, some items specific to your system might not be herein documented.

Note

This information is also applicable to customers using v8.33 software.

3.1 Mechanical Health

3.1.1 Reducer Drive Health

Although FANUC reducer drives have a substantial design life and high reliability, certain application conditions can degrade the drive life including; harsh environments, inadequate lubrication, and accidental equipment collisions. ZDT monitors the drive test data and can determine if a drive is degrading, long before any external signs are visible.

3.1.1.1 Reducer Drive Health Setup

The Reducer Drive Health feature requires the execution of a Robot Condition or self-test program to be executed on a regular basis. Paint robots should have their self-test program run at least once a week. For other robots it is recommended to run daily or once per shift.

This ZDT health suite detects abnormal reducer conditions to alert you to a potential impending failure before downtime occurs.

- Requires creation of a diagnostic program (or self-test), mastering of initial data, and periodic execution of the diagnostic program.
- You automatically create the diagnostic program by recording a base position and setting the allowable motion range. +/- 10 degrees of motion is required for proper analysis.
- Run the diagnostic program periodically or ideally once a day/shift. The more you run the program, the better the chance of predicting a failure.

3.1.1.2 Creating the Robot Condition Program

About this task

Use this procedure to create the robot condition program.

Before you begin

- Make sure that your payload schedule is correct and that the measurements are accurate.
- For paint systems with three or more motion groups, you must isolate the opener robots. The condition test is only for painting robots.

Procedure

1. Jog the robot to the base position, where the diagnostic program starts and ends.
2. Press **MENU**.
3. Select **Utilities**.
4. Press **F1**, [**TYPE**].
5. Select **Robot Condition**.

You will see a screen similar to the following.

The screenshot shows the 'Robot Condition' screen with the following data:

Detection Program Creation				1/10
Base Program Name:				
DIAG				
Group Number:				1
Payload Schedule:				1
Base Position: (recorded)				
Motion Range				
	Min(rel)	Base	Max(rel)	
J1:	- 10.000<	0.000>+	10.000	
J2:	- 10.000<	0.000>+	10.000	
J3:	- 10.000<	0.000>+	10.000	
J4:	- 10.000<	0.000>+	10.000	

Figure 3.1.1.2 Robot Condition Screen

6. Move the cursor to **Base Program Name**, and type the name of the diagnostic program that you would like to create.
The group number will be added to the base name to create the program name. The base program name should not be more than 34 letters. Typically, **DIAGx** is used where x is the group number.
7. Move the cursor to **Group Number**, and type the number of the group for analysis.
8. Establish a base position of the robot.
 - a) Enter T1/T2 or Teach mode and jog the robot to a position in the cell that will allow each axis to move roughly +/- 10 degrees.
Carefully determine this base location. Jog the robot to a safe position that facilitates collision free motion (within a limited range).

- b) Move the cursor to **Base Position** and record the current robot position as the base position.

You must be in T1/T2 or Teach Mode. Press and hold **SHIFT** then **F4, RECORD** to record the base position.

The software will automatically create the condition program from a base location defined by the user.

Note

The robot will move roughly +/- 10 degrees for each axis during motion. Motion during this program must be collision-free and not damage robot, tooling, or dress. Often the Base Location can be Home or a maintenance position, if they meet the above criteria.

9. Set the **Payload Schedule** to the correct payload.

10. Enter the allowable motion range of each axis in degrees.

Note

The robot will move roughly +/- 10 degrees for each axis during motion. Motion during this program must be collision-free and not damage robot, tooling, or dress. Often the Base Location can be Home or a maintenance position, if they meet the above criteria.

11. Disable the teach pendant and switch to T2 or AUTO mode (Paint systems would be Manual or Production mode) and reset all faults.

12. Set the override to 100%.

13. Press **SHIFT** and **F2, CREATE**.

A program will be created with group number added to the **Base Program Name**.

It is normal for the robot to fault when program creation is complete.

14. Run the program just created (in this case DIAG1).

Run with low override to assure safety of the program in order to verify that the robot clears all obstructions and then run the program at 100% to verify operation

! CAUTION

Make sure that the robot path is clear of the paths of adjacent robots to avoid collisions.

15. Adjust the path, if unable to reach all positions.

Adjust the limits and recreate program again.

Note

If DCS I/O Connect is the only thing used, please BLOCK THE LIGHT SCREEN WHILE TESTING THE PATH.

The program might need to be adjusted if the path interferes with DCS Zones: If the robot is loaded with DCS Pos Speed Check, verify that the DIAG1 path can run without posting an Unsafe condition (or breaking the light curtain while running).

16. If an `Unsafe` condition occurs, make appropriate adjustments on the affected axis.

Re-teach base position, limit amount of motion to one direction or the other (example: `Min = -20 / Max = +0`) based on where the violation occurs.

17. If the diagnostic path (`DIAG1`) causes collisions or disturbance excess warnings due to aggressive motion, adjust the following to relieve the condition.

- a) Re-teach the base position to relieve the affected axis (e.g. J3 is ~90 degrees from the floor. In this case adjust the angle to be lower when teaching the base position.)
- b) Adjust the speeds in the Robot Condition Path.

3.1.1.3 Run the Robot Condition Program

Once the program has been verified (no alarms, no collisions), run the program several times in Auto or T2 at 100% override speed.

For PaintTool systems, it is suggested to use `job97` for group1 and `job96` for group 2 to execute the self-test program, although if they are in use by the plant, any job number can be used.

This job will be executed on all robots, on all controllers at the same time. Therefore, it is important to verify that all robots can move from HOME to the baseline position and back to the HOME position without colliding with any other robots in the zone.

3.1.1.4 Master the Robot Condition Baseline

It is necessary to establish the baseline after the initial teaching of the self-test. It is also necessary to re-establish the baseline after a reducer replacement.

CAUTION

This should only be performed after running the robot at 100% Override Speed in Auto (Paint systems use Production or Manual) or T2 mode (i.e. the robot condition path has been verified).

Inappropriate baseline values will affect the sensitivity of ZDT analytics. When establishing the baseline please check the following:

- Self-test can run free from all obstacles and has no dressout snagging
- Robot and reducers have no visible or obvious damage
- Robot and end effectors are mounted securely
- There is no major grease leakage on/around the robot
- The robot is not mis-positioning (if it is, check axis backlash)

If you find (and correct) an issue, reset the baseline value as described in this section.

The following items may prevent this function from working properly:

- Editing the created program
- Creating the program with the same name will overwrite the old program. If overwritten after MASTER, this function will not work properly. If the program needs to be changed during setup, go through both CREATE and MASTER.
- Different programs may be created with arbitrary base positions. However, an axis can be analyzed with only one program. With different programs analyzing the same axis, this function will not work properly due to different characteristics of the data collected. When trying to analyze an axis with a program different from the one used for MASTER, a warning will be posted (different axes can be analyzed with different programs).

- Basically, go through MASTER only with initial setup and when replacing a reducer.
- If it becomes impossible to use the program for some reason, such as moving the robot to different location, it would be necessary to go through CREATE and MASTER. However, if there is a reducer already reaching an end of its operating life, then going through these steps may cause this function not to work properly.

3.1.1.4.1 Mastering the Robot Condition Baseline

About this task

Use this procedure to establish or re-establish the baseline.

Before you begin

Robot condition path has been verified.

Procedure

1. Run the program five times at 100% prior to setting the baseline.
This will warm up the robot and give more accurate results.
2. Navigate to the menu to establish the baseline data.
MENU > 0 > STATUS > F1, [TYPE] > Robot Condition.
3. If the upper right display of GRP [] does not show the proper group for analysis, press **F2, GROUP** and enter the group number.
4. Press **F4, [MASTER]**.
5. If all axes (typical) need to be Mastered/Baseline Set then type in -1 and press **[ENTER]**.
6. If only one axis needs to be Mastered/Baseline Set, then only select the axis to be set #, **[ENTER]** (# = 1 – 6).

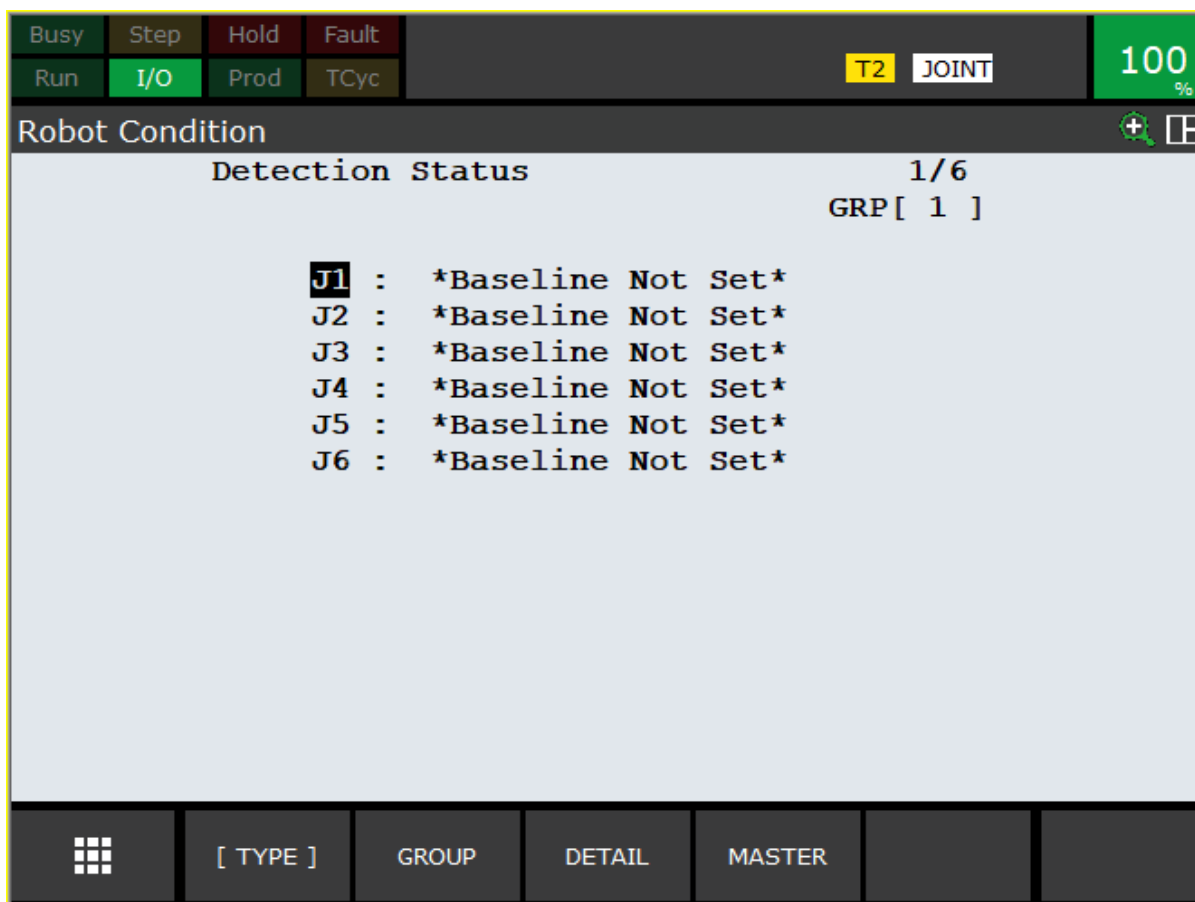


Figure 3.1.1.4.1 Robot Condition Detection Status Screen

3.1.1.5 Setup the execution of the Robot Condition Program

For optimum results, the more often the robot condition program is executed, the better chance the analytics have to detect an impending problem.

Systems with no cell control (PLC).

On systems without a PLC, please run the self-test program at least once per day at 100% speed override. Please make sure that the actual payload is matched with the payload schedule called in the self-test program.

Systems with a cell control

Modify the Cell PLC so that the self-test program can be run once per day. Some PLCs use a counter, others provide a notification to the user that the self-test is required. Typically at the start of the shift before the first job enters the robot envelop the PLC will request the self-test to run.

3.1.2 Excessive Sudden Stops

Also known as Flight Data Recorder, this function monitors all sudden stops (such as e-stops, collisions, etc.) that may occur at any time for all connected robots. Many times, users are not aware of the amount of sudden stops occurring during production. These stops not only cause production interruptions, but may also reduce the drive life of the robot, especially if the stops happen in the middle of high speed, high torque moves by the robot. Because ZDT alerts you to excessive sudden stops, you can be proactive in correcting the issue before downtime occurs, or extra maintenance is required.

Flight Data Recorder must be enabled from the **SETUP ZDT Client** Screen with *iR*Diagnostics. See [Section 2.1, ZDT Client Setup on the TP](#).

3.1.3 Job Data (High Water Mark Job Data)

FANUC robot drive systems have high reliability and consistent performance. Because ZDT can analyze the torque profile of each axis during job execution, variations are detected and will alert you of any potential changes to motor health or drive health before any visible signs are indicated in the application process. ZDT will alert you of issues in advance, so maintenance can be scheduled before incurring production downtime. The cost of a motor/drive failure in production can be quite large.

By default, for most job running conditions, Job Data, or High Water Mark (HWM) Job Data, monitors the TP program running status and determines the timing to start/stop tracking Job Data and then send the data to the data collector. No extra setup is required. For example, the PLC runs a style program each time; then Job Data will start when a style program starts and the data will be sent when a style program ends. However, for multi-tasking programs, or continuously looping programs, some extra setup is required. See [Section 3.1.3.1, Mode Selection](#).

3.1.3.1 Mode Selection

This section is only relevant if your system uses multi-tasking programs, or continuously looping programs.

For continuously looping programs, please select the appropriate Job Data triggering mode in ZDT Client's **iR**Diagnostics Messages menu by moving the cursor to **5 Mode** and press **[CHOICE]**.

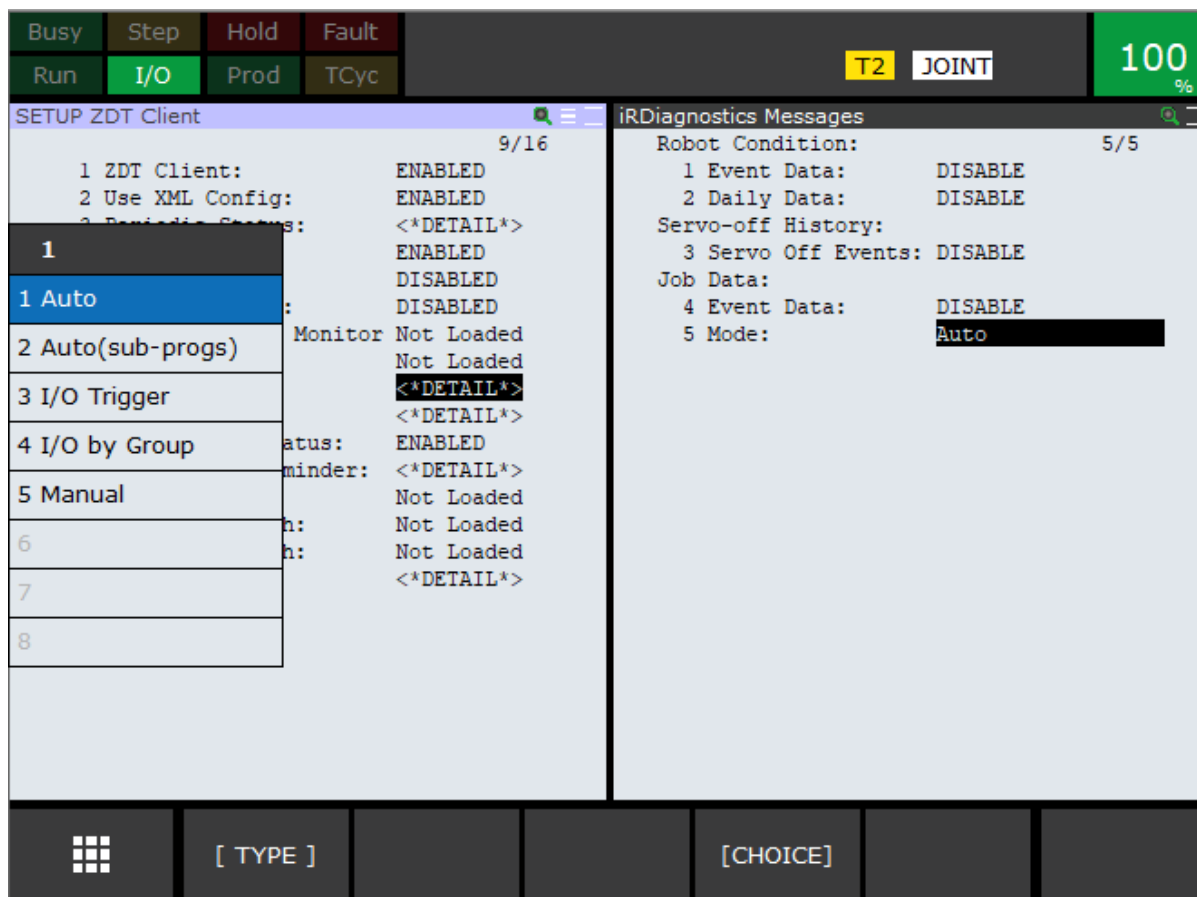


Figure 3.1.3.1 (a) Job Data Mode Selection

Job Data Triggering Modes:

1. **Auto:** This is the default triggering mode. HWM Job Data is recorded when any TP program starts running and data is sent when the TP program ends.
2. **Auto(sub-progs):** This is for continuously looping programs. Selecting this mode, HWM Job Data is recorded when any sub TP program (CALLED program) with motion lines is called. Data is sent when the sub TP program returns back to the main program.
3. **I/O Trigger:** An I/O point is used to trigger the Job Data. I/O type, port number, and trigger type (when the I/O turns on or off) can be set by using the menu items shown in [Figure 3.1.3.1 \(b\)](#).

Also, the job name can be specified in this triggering mode.

If **Job name to use** menu item is MANUAL, the value of the **Job name to use** I/O point at the time when HWM Job Data is sent will be attached to the end of the “Job” string as a job name. For example, if the following items are set:

Job name to use:

MANUAL

I/O Type

AI

I/O Port

3

and AI[3] = 2 at the time when HWM Job Data is sent, then job2 will be used as the job name.

If the **Job name to use** menu item is STYLE, then the name specified in the Style table will be used as the job name and the index of the Style table will be specified by the I/O point. For example, if the following items are set:

Job name to use:

STYLE

I/O Type

GI

I/O Port

5

and GI[5] = 1 at the time when HWM Job Data is sent, then the style name in the Style table index 1 will be used as the job name.



Figure 3.1.3.1 (b) I/O Triggering setup screen

4. **I/O by Group:** this mode supports different I/O triggers per each motion group. Please contact FANUC for details.
5. **Manual:** this mode supports manual triggers by inserting system variables. Please contact FANUC for details.

3.1.4 Flexible Data

Flex Data and Flex I/O are robot controller features that provide ZDT users the ability to add application-specific or process-specific cycle-based information to cloud data for their facility. Items such as KAREL variables, system variables, registers, and I/O ports are configured on the controller. When the data reaches the cloud, it is combined with program name, cycle time and power consumption data and is available for viewing and exporting on the **Cycle Status data view** page on the ZDT web portal.

3.1.4.1 Installation and Configuration

The recommended configuration method is to create a CF file that sets the system variables described in this document. This way multiple controllers can be setup easily. CF files are simple text files with a .cf extension that can be used to set system variables. To execute a CF file on the robot, see the following procedure.

3.1.4.2 Flex Data Configuration

Flex data supports up to 10 different fields. The value sent by flex data is configurable. It can be configured to send the value when the job ends, or configured to send the highest value during the job.

The following example shows how a CF file could setup collection of three pieces of data; R[1], the system variable *\$fast_clock*, and a KAREL variable (*myprogvar*) from the KAREL program *myprog*.

```
SET VAR $condet_cfg.$ext_data[1].$reg_num = 1
SET VAR $condet_cfg.$ext_data[1].$style = 2
SET VAR $condet_cfg.$ext_data[2].$prog_name = '*system*'
SET VAR $condet_cfg.$ext_data[2].$var_name = '$fast_clock'
SET VAR $condet_cfg.$ext_data[2].$style = 2
SET VAR $condet_cfg.$ext_data[3].$prog_name = 'myprog'
SET VAR $condet_cfg.$ext_data[3].$var_name = 'myprogvar'
SET VAR $condet_cfg.$ext_data[3].$style = 2
SET VAR $condet_cfg.$ext_mask = 1023
SET VAR $condet_cfg.$enable = 1
```

3.1.4.2.1 Triggering Instructions

When using software prior to V8.30P/43 for R-30iB, V9.10P/13 for R-30iB Plus, the data-gathering mask must be reset manually after each job. This is done by setting the variable *\$condet_cfg.\$ext_used* = 0. Use a TP program to set the data mask. Depending on which method of triggering is in use, there are slightly different approaches to the setting of this variable. Triggering mode is defined by *\$condet_cfg.\$mode*.

When using **auto trigger mode**, reset the mask after the job program completes. The job program must complete first. This can be accomplished in a few different ways.

1. The recommended method is to use the PLC to run a TP program that sets the variable *\$condet_cfg.\$ext_used* = 0. This method is recommended to eliminate the possibility that the variable is set before the job program completed.
2. Alternatively, if using UOPs, another method is to create a background logic TP program similar to [Figure 3.1.4.2.1 \(a\)](#). This logic increases a register value each time the BG program runs when a job is not running, and sets the variable approximately 400 ms later (R[100] * ITP time). This time can be adjusted, if needed, but should be at least 100ms and no longer than it takes for the PLC to initiate the next program.

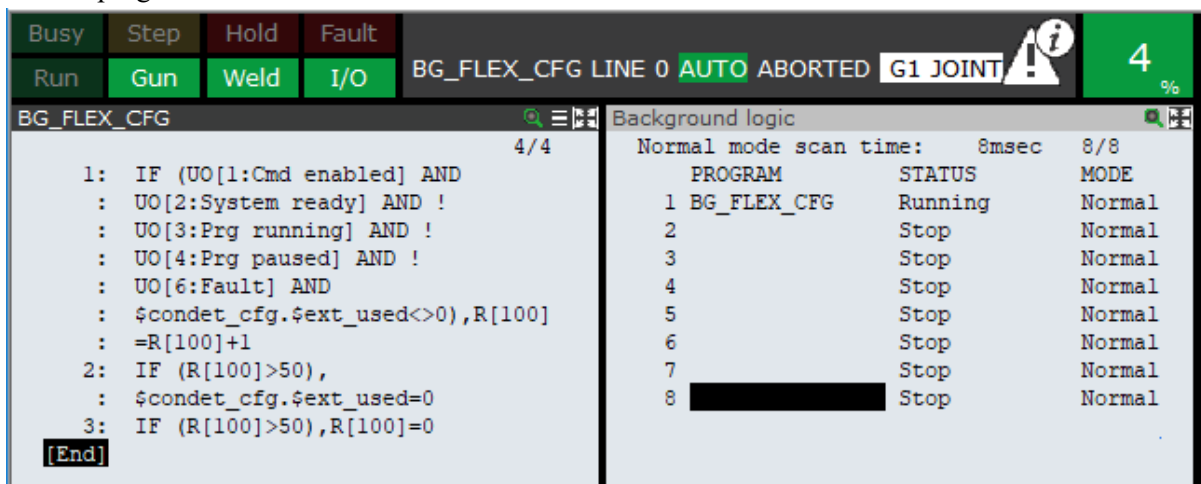


Figure 3.1.4.2.1 (a) Background program to reset the mask

3. Via a TP macro with an I/O trigger (Press **MENU** > **SETUP** > **Macro**). In the TP macro, set *\$condet_cfg.\$ext_used* = 0. The I/O trigger to run the macro can be any source available on the **Macro Command** screen. It is very important that the variable not get set until the job program completes. The I/O trigger must be reset after use and can be done in the same program that sets the variable. See [Figure 3.1.4.2.1 \(b\)](#).

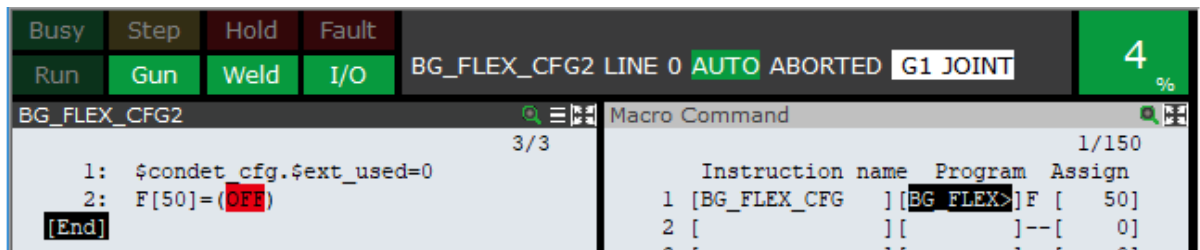


Figure 3.1.4.2.1 (b) Macro to reset the mask

4. If using the Home I/O feature, set the variable in the Home I/O program.

When using a **manual trigger mode** (Sys or I/O), the mask can be reset in the job program before the manual trigger is activated. Add a line to the TP program that sets the variable `$condet_cfg.$ext_used=0`.

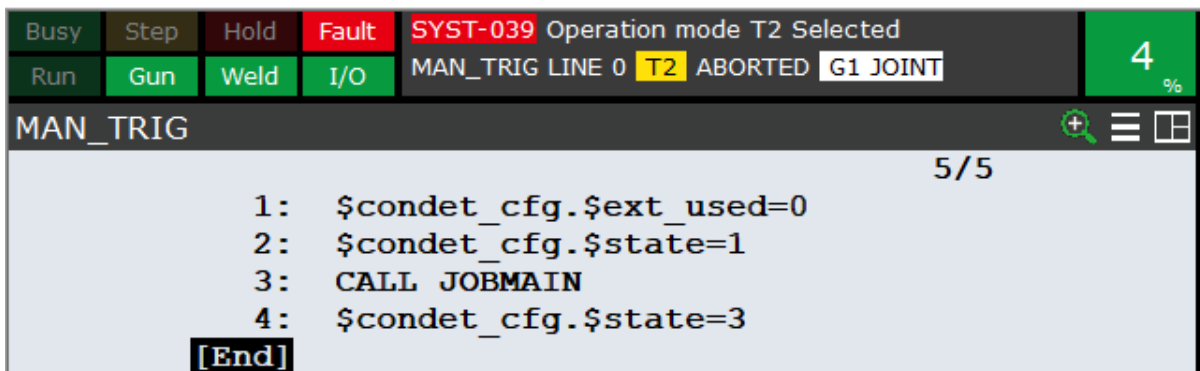


Figure 3.1.4.2.1 (c) Resetting the mask when using manual trigger mode

3.1.4.3 Flex IO Configuration

Flex I/O is used to identify changes in I/O states or values during a job.

It has four different types of detection styles. The detection styles identify if the I/O.

1. Turned off during the job
2. Turned on during the job
3. Went over a limit during the job
4. Went under a limit during the job

Table 3.1.4.3 I/O Types

Value	Description
1	Digital input
2	Digital output
3	Analog input
4	Analog output
5	Tool output
6	PLC input
7	PLC output
8	Robot digital input
9	Robot digital input
11	Opertor panel input

Value	Description
12	Operator panel output
16	Weld inputs
17	Weld outputs
18	Group inputs
19	Group outputs
20	User operator panel input
21	User operator panel output

The following example shows how a CF file could be used to setup flex I/O to identify when:

- DO[5] turned off
- AI[2] exceeded 25

```
SET VAR $condet_io.$io_data[1].$io_type = 2
```

```
SET VAR $condet_io.$io_data[1].$io_port = 5
```

```
SET VAR $condet_io.$io_data[1].$style = 1
```

```
SET VAR $condet_io.$io_data[2].$io_type = 3
```

```
SET VAR $condet_io.$io_data[2].$io_port = 2
```

```
SET VAR $condet_io.$io_data[2].$style = 3
```

```
SET VAR $condet_io.$io_data[2].$limit = 25
```

```
SET VAR $condet_io.$req_mask = 31
```

```
SET VAR $condet_io.$enable = 1
```

3.1.4.4 Advanced Configurations

It is possible to modify which ten pieces of Flex data are collected during each job. For example, one job can use flex data one through five, and another job could use six through ten. To accomplish this, a variable needs to be modified before each job. This variable is a bitmask for each field, where each bit is a flag for the data point to be sent.

The field `$condet_cfg.$ext_mask` is used to modify which flex data points are sent.

Setting the variable as indicated below sends data points 1-5 for job 1 and data points 6-10 for job 2.

- For job 1 set `$condet_cfg.$ext_mask = 31` (1 + 2 + 4 + 8 + 16) (Binary: 00 0001 1111)
- For job 2 set `$condet_cfg.$ext_mask = 992` (32 + 64 + 128 + 256 + 512) (Binary: 11 1110 0000)

Flex I/O cannot be modified to send different items without a power cycle.

3.2 System Health

3.2.1 Controller Battery Low

The robot contains a battery that supplies power to the axis encoder in the event of a power failure. Since the batteries have a usable life, they must be changed periodically. ZDT monitors the battery status and can alert you of a low battery condition in advance so that maintenance can be scheduled around

production. If the battery dies and power to the robot is lost, it can take from several minutes to several hours (depending on the application and the number of robots) to re-master and resume production.

ZDT will alert you when a low battery condition exists so that maintenance can be performed before power is lost. The cost of having to re-master a robot in production can be quite large.

No setup is required for this feature.

3.2.2 TPP Program Changes

Data collected is sent to the data center where Teach Pendant Program (TPP) changes can produce an alert.

Many users have experienced a time when their production operation suddenly changes and doesn't perform as expected. Many times this is caused when a user changes a program in the robot controller. ZDT monitors and alerts the customer to any changes that occur. Because some changes have a detrimental effect on the production operation, having knowledge of the change may prevent the negative impact that the change may produce. This may save production losses and downtime.

You need to enable this function in the ZDT Client screen in order to track the program change. See [Chapter 2, ZDT COMMUNICATION AND BASIC SETUP](#).

3.2.3 Controller Memory Monitor

The robot controller utilizes multiple memory types. If memory levels were to go low, the system could crash or not function properly. ZDT monitors the memory levels and alerts you if a low memory level condition exists or is predicted to occur soon by comparing daily values to warning and critical limits.

ZDT monitors these memory types:

- *TEMP*: DRAM. This can affect system operations if low. The warning limit is 1 MB and the critical limit is 0.3 MB
- *MBEM (MAIN)*: This is used to receive data from MAIN. This can block the communication between MAIN and COMM processors if low. The warning limit is 3 KB and the critical limit is 2 KB.
- *MBEC (COMM)*: This is used to receive data from COMM. This can block the communication between MAIN and COMM processors if low. The warning limit is 3 KB and the critical limit is 2 KB.
- *PERM*: System memory. This can affect system operations if low. The warning limit is 40 KB and the critical limit is 15 KB.
- *TPP*: Teach Pendant Program memory. This can block new programs from being loaded if low. The warning limit is 10 KB and the critical limit is 5 KB.
- *FROM*: Flash memory. This can block new features from being loaded if low. The warning limit is 0.4 MB and the critical limit is 0.15 MB.

ZDT will alert you in advance of a low memory level condition and provide recommended actions so that maintenance can be performed before a system crash or other negative effect can occur.

No setup is required for this feature.

3.2.4 Variable Monitor Setup

This section is included for reference. The ZDT application will provide its own configuration file for the application you are using.

The ZDT variable change monitor is used to scan certain variables and I/O and notify you when there is a change. The list of items to be monitored is defined in an XML file. On startup, this file is parsed and the monitor structure is built. A scan utility is called periodically to monitor changes to the listed items. When one is changed, a message describing the change is queued. ZDT provides a queue timer and queue size to send a message as to whichever occurs first.

The types of items that can be monitored include:

- System Variables
- Numeric Registers
- String Registers
- Position Registers
- Frames
- Program Variables
- I/O

3.2.4.1 Configuration

Setup is required to use this service. In the **SETUP ZDT Client** menu, **Var Change Log** must be enabled along with **ZDT Client**.

Secondly, `VARCHG.XML` must be configured. This configuration file defines the items to be monitored. Create `VARCHG.XML` offline and load it using the procedure defined below. This file will be backed up and restored. Each time you modify `FR:\VARCHG.XML`, you must cycle power for the changes to take effect.

The tag definitions for the `VARCHG.XML` file are described in [Table 3.2.4.1](#).

Table 3.2.4.1 Tag definitions for the `VARCHG.XML` file

Tag	Parent Tag	Depth	Definition
<CHANGELOG>	None	0	This is the root tag

Tag	Parent Tag	Depth	Definition
<VAR>	<CHANGELOG>	1	This tag defines the variable and contains following attributes: <i>var</i> - This attribute is REQUIRED if the item is a variable. It is the system-defined name of the variable <i>io_type</i> - This attribute is REQUIRED if the item is I/O. It describes the type of I/O to monitor (Digital IN, Digital OUT, Analog IN, etc.). <i>io_index</i> - This attribute is REQUIRED if the item is I/O. It is the index of the I/O to monitor. <i>program</i> - This attribute is OPTIONAL if the VARCHG tag is describing a system variable, position register, numeric register, or string register. Otherwise it is REQUIRED if the item is a variable. This is the name of the program that the variable belongs to. <i>title</i> — This attribute is OPTIONAL. It is used to define the title, a user defined description of the variable. The default title is [prog_name]var_name. <i>dict</i> - This attribute is OPTIONAL. It is used, along with ele_no, to define a dictionary entry. This entry can be used instead of title to describe the title in multiple languages. You must omit the title attribute to use a dictionary entry. <i>ele_no</i> - This attribute is OPTIONAL. It is an integer. See the description for dict. <i>subtype</i> - This attribute is OPTIONAL. It is used to define the subtype, which is used to classify the change notification in the ZDT application. The default subtype is the title. <i>subt_dict</i> - This attribute is OPTIONAL. It is used along with subt_ele to define a dictionary entry. This entry can be used instead of subtype to describe the subtype in multiple languages. You must omit the subtype attribute to use a dictionary entry. <i>subt_ele</i> - This attribute is OPTIONAL. It is an integer. See the description for subt_dict. <i>alert</i> - This attribute is OPTIONAL. It can be either 0 or 1. It is used to describe the alert status in the ZDT application. Unlike other messages, alert does not immediately send the change, nor send an alert text message to the phone. It defaults to 0.

Tag	Parent Tag	Depth	Definition
<ENUM>	<VAR>	2	This tag allows you to display the user definition of the value. It has two attributes, value and string, to provide the mapping.
<EXCLUDE>	<VAR>	2	Certain fields can be excluded from structure or arrays using this configuration. It has a field to list the exclusion variable.

The following shows a sample configuration file.

```
<?xml version="1.0" ?>
<CHANGELOG>
  <VARCHG prog="my_prog" var="my_var" />
  <VARCHG prog="PKMAIN" var="parts_picked"
    dict="PICK ele_no="12"
    subt_dict="PICK" subt_ele="11"/>
  <VARCHG var="$POSREG[1,1]" title="Job 1 Positions"
    subtype="Job Vars" alert="1"/>
  <VARCHG var=$NUMREG[1] dict=XOPT ele_no="6"/>
  <VARCHG var=$STRREG[1] dict=XOPT ele_no="7"/
    subt_dict="XOPT" subt_ele="20"/>
  <VARCHG var="$UI_CONFIG.$TIMEOUT"
    title="iPendant Time Out"
    subtype="UI Configuration"/>
  <VARCHG var="$UI_CONFIG.$READONLY[2]"
    title="Navigate iPendant RW Access"
    subtype="UI Configuration"/>
  <VSRCHG io_type="DIN" io_index="1"/>
  <VARCHG io_index="2" io_type="DOUT"/>
</CHANGELOG>
```

3.2.4.2 IO Type Definitions

To monitor an I/O value, you need both a type and an index. The index is the port number. The type must be one of the pre-defined constants described below.

- Digital Input: DIN
- Digital Output: DOUT
- Analog Input: ANIN
- Analog Output: ANOUT
- Group Input: GPIN
- Group Output: GPOUT
- PLC Input: PLCIN
- PLC Output: PLCOUT
- Robot Input: RI
- Robot Output: RO
- Operator Panel Input: SOPIN
- Operator Panel Output: SOPOUT
- Teach Pendant Input: TPIN
- Teach Pendant Output: TPOUT
- Weld Input: WI
- Weld Output: WO
- User Operator's Panel Input: UOPIN

- User Operator's Panel Output: UOPOUT
- Laser Input: LDIN
- Laser Output: LDOUT
- Weld Stick Input: WSIN
- Weld Stick Output: WSOUT
- Laser Analog Input: LANIN
- Laser Analog Output: LANOUT
- Tool Output: TOOL
- Brake Output: BRAKE
- Emergency Stop: ESTOP

3.2.4.3 Excludes

This service includes the ability to exclude monitoring particular elements of an array or a complex data structure. This can be useful in certain situations, as when you want to monitor a data structure, but a non-informative field in that structure is constantly changing. To configure, an `<EXCLUDE>` tag is needed. `<EXCLUDE>` tags are children of a `<VARCHG>` tag.

Each `<EXCLUDE>` tag has one attribute:

field - This attribute is REQUIRED. It is the system-defined name of the excluded element. The definition of field must be some sub-element of the variable described in the parent `<VARCHG>` tag. If the variable is an array of structure, then the fields are excluded from all structures in the array.

An example of how this might be used is shown in [Figure 3.2.4.3](#).

```
<?xml version="1.0" ?>
<CHANGELOG>
  <VARCHG var="$STYLE_NAME">
    <EXCLUDE field="$STYLE_NAME[2]" />
    <EXCLUDE field="$STYLE_NAME[3]" />
    <EXCLUDE field="$STYLE_NAME[9]" />
    <EXCLUDE field="$STYLE_NAME[10]" />
  </VARCHG>
</CHANGELOG>
```

Figure 3.2.4.3 Use of field Attribute

In this example, `$STYLE_NAME` is an array with at least ten entries. The user has configured the change monitor so that any changes to entries [2], [3], [9], and [10] of this array will not be detected by the change monitor; it will completely ignore them.

Note

`<EXCLUDE>` tags can only be used on arrays or complex variables. These are not supported in registers including POSITION, NUMERIC, STRING, or VISION registers.

3.2.4.4 Enumerated Values

Getting a number back as a value can sometimes be uninformative. Perhaps the values 0 and 1 really correspond to Hand Closed and Hand Opened. To clarify the return values, the change monitor lets you set up enumerated values. These numerical values correspond to a string. When defined, ZDT will display this defined string in any returned output as a substitute for the numerical value. To configure, an `<ENUM>` tag is needed. `<ENUM>` tags are children of a `<VARCHG>` tag.

Each `<ENUM>` tag has two attributes:

- *value* - This attribute is REQUIRED. It is the numerical value you want replaced.

- *string* - This attribute is REQUIRED. It is the text which will replace the numerical value for any output from ZDT.

An example of how you might use this is shown below.

```
<?xml version="1.0" ?>
<CHANGELOG>
  <VARCHG var="$ACC_MINLMT"
    ENUM value="1" string"Minimum limit of ACC is 1."/>
    ENUM value="2" string"Minimum limit of ACC is 2."/>
  </VARCHG>
</CHANGELOG>
```

In this example, when *\$ACC_MINLMT* is equal to 1, ZDT will return the text *Minimum limit of ACC is 1.* instead of the value 1. The same happens for value 2 with the defined string for 2. However, if *\$ACC_MINLMT* is any other value, the numerical value is returned.

3.2.4.5 String Replacement

As described above, several configuration parameters require a string of text. The change monitor allows you to use a variable to define this string, instead of a pre-defined string. Unlike the pre-defined string demonstrated above, strings defined through a variable are not constant. The change monitor will look at the current value of the variable and use that as the text. The change monitor allows System Variables, KAREL Variables, and String Registers to be used instead of a string. The following configuration would be valid:

```
<?xml version="1.0" ?>
<CHANGELOG>
  <VARCHG var="$ACC_MINLMT" title="APPLICATION[1]">
    ENUM value="1" string="$STRREG[1]" />
    ENUM value="2" string="[my_k_prog]my_k_str" />
  </VARCHG>
</CHANGELOG>
```

In this example, title is defined as the current content of *\$APPLICATION[1]*, a system variable that is a string. The first ENUM string is defined as *\$STRREG[1]*, or the current content of the first string register. The second ENUM is the current content of the KAREL variable *my_k_str*, defined in the KAREL program *my_k_prog*.

IMPORTANT: This functionality assumes a STRING variable is used to define content. If a numerical value is used, the system will attempt to interpret it as a string, resulting in a nonsense string.

3.2.4.6 Change Monitor Output

Changes made to the monitored variables and I/O are queued and periodically sent to the ZDT Data Collector. See Queue Timer and Queue Size, in the ZDT screen. The ZDT Data Collector will display the changes, including new values of the variables and I/O and, in most cases, the old values. The variables and I/O are scanned every 60 seconds by default. To change the scan time, set *\$RCMCFG.\$VARCHG_TIME* from the System Variables menu. The minimum scan time is 15 seconds.

Note

When structure is being monitored, then only the changed field is reported.

3.3 Process Health

3.3.1 Application Data Monitoring

Note

Currently not available with PaintTool.

ZDT monitors many production related factors of your application including; running, idle, and faulted. It's the equivalent of having an on-site industrial engineer performing a time study on your robotic application. This important data will allow you to understand what factors are key to gaining production throughput and improving the overall application process.

This ZDT function may allow you to make gains in overall production rates, eliminate bottlenecks, and improve efficiency by potentially spreading work over more equipment.

Please visit the ZDT Client screen described in [Chapter 2, ZDT COMMUNICATION AND BASIC SETUP](#) for details as to how to setup the Application Data Monitoring function.

3.3.2 ServoGun Monitoring ZDT

This function monitors and diagnoses the disturbance torque during servo gun close motion.

This health suite is only applicable if your robot is configured with a servo gun connected to the FANUC controller and ServoGun software is installed.

Robots, with a servo gun application, test the mechanical integrity of the gun **during tip wear measurement**. ZDT monitors the results of the test and it will alert you if the servo gun is out of specification. In addition it will show data trends. An alert is provided well in advance of any noticeable degradation of the servo gun.

This ZDT function identifies when a significant change has occurred in the motor's response at gun close. This is an indication that tip pressure may be out of tolerance. It can also indicate that mechanical damage has occurred.

Data is collected during TIP wear update (when program wr_upd01 or tw_upd01 run, for example). ServoGun ZDT is designed to capture measurement data at 10:00 am each day (according to the robot's clock).

3.3.2.1 Setting Torque Tolerance

About this task

Use this procedure to set the torque tolerance.

Before you begin

Servo gun is used.

Note

If the ServoGun diagnostic menu does not appear, it may be disabled. Please set the following system variable: `$SGZDTSYS.$SCRN_MASK = 3`

Procedure

1. Press **MENU**.
2. Select **SETUP**.
3. Choose **Servo Gun**.
4. With the cursor on **General Setup**, press **ENTER**.
5. Cursor to **Servogun diagnosis** and press **ENTER**.
6. Change servogun **Max Torque Tolerance** and **Average Torque Tolerance** settings.
7. Go to **SETUP Servo Gun Diagnosis: Gun > General Setup > Servo Gun Diagnosis**.
8. Confirm the settings in [Table 3.3.2.1](#).

Table 3.3.2.1 Servo Gun Diagnosis Values

SERVO GUN DIAGNOSIS	Value
Max Torque Tolerance	100%
Average Torque Tolerance	100%

3.3.3 Paint EStat Cable Monitor Systems PaintWorks Required

This item is only available on Paint systems using FANUC PaintTool control of electrostatics. FANUC e-stat cables are designed for high reliability, but factors such as overly aggressive motion and excessive exposure to solvents can reduce the cable life. ZDT monitors the cable performance and can determine in advance if the cable is showing signs of degradation. If the cable performance decreases enough, production can be interrupted or serious quality issues could result in re-work or scrap vehicles.

ZDT will alert you of issues in advance so that you can schedule maintenance before incurring quality issues or production downtime. The cost of an e-stat cable failure in production can be quite large.

Estat Max Current (µa)

The controller monitors the current feedback to insure that a minimum current (µa) is achieved during a normal production job. You can adjust the value monitored by setting the following variable:

- Program: PAVRZDT
- Variable: *EstatZDTCMOS[group].iMinThres*
- Default: 7

To prevent false alerts while running ghost jobs without sheet metal, the PLC should set robot controller DIN[366] to ON during ghost jobs.

3.3.4 Paint Rail Torque Monitor Systems PaintWorks Required

This item is only available on Paint systems using FANUC PaintTool control.

FANUC rails have high reliability but excess paint build-up, or bearing and cam follower wear can produce torque faults. ZDT monitors the torque values during production and will alert you of higher trending torque data well ahead of incurring a fault which will interrupt the paint process, causing downtime and potential quality issues.

ZDT will alert you of paint rail issues in advance so that maintenance can be scheduled before incurring production downtime. The cost of a torque fault in production can be substantial.

3.3.5 Paint Process Regulator Health SMC ITV Regulators Systems PaintWorks Required

The process regulator controls the painting parameters such as Bell Speed and Shaping Air. Any performance issues with the device may have an immediate impact on paint quality. ZDT monitors the results of manually run regulator tests. ZDT will alert you well in advance of potential performance issues of the regulator that might affect paint quality.

ZDT will alert you of issues in advance so that maintenance can be scheduled before incurring quality issues or production downtime. The cost of a regulator failure in production can be quite large.

3.3.5.1 Regulator Health Setup

3.3.5.1.1 ITV Test

What does the ITV test perform?

- It checks the integrity of the ITV at the low end of the operating range, which is a good indicator of worn solenoids in the ITV.
- The test covers both Bell Speed and Shaping Air ITV's, as well as bell speed fiber optics.
- It runs in closed loop mode through four set points and assures that the set point is reached (threshold).
- It runs in open loop mode at zero output to detect upward movement of the output (creep).
- If the test fails, the system may still function but should be checked.

3.3.5.1.2 Bell Speed and Shape Air Regulator Test

This test is executed by running a ghost job (typically `style 98`) from the PaintWorks GUI. This ghost job should be executed via the ZDT option on the PaintWorks Booth Status screen.

This test does not move the robot, although the robot may perform a color change as it is necessary for the applicator to be enabled while running the test. After the job is complete, the PaintWorks system updates the status of the test on the ZDT alert screen.

The ITV test is designed to check for clogged exhaust, leaking internal solenoids or leaks in general. The test involves stepping through a range of command values at the low or near closed range, to insure the aforementioned failures are detected.

By default, the following job TP program is loaded per painter group with the Versabell option:

```
ZDTITVPX.TP  
Where X = Group
```

The above job(s) must be renamed or copied to a job number that is not in use by the plant. By default it is recommended to use `job 98`, if available.

When running the job in automatic, it is required that the applicator is enabled.

It is recommended that this test is executed weekly, although it can be executed as frequently as desired.

3.3.6 P700iA Gear Drive Backlash Detection Systems PaintWorks Required

Although FANUC gear drives have a substantial design life and high reliability, application conditions can degrade the drive life including; harsh environment, inadequate lubrication, and accidental equipment collisions. ZDT monitors the drive test data and can determine if a drive is degrading, long before any external signs are visible. ZDT will alert you of gear drive issues in advance so maintenance can be scheduled before incurring production downtime. The cost of a gear drive failure in production can be quite large.

3.3.6.1 J1 Gear Backlash Detection Health Check Setup

The J1 Backlash Test checks the integrity of the J1 drive regarding backlash between the pinion and ring gear. To determine if the results are within specified limits, a special motion program is executed and analyzed at the end of the test. This test is only valid for the P-700iA Robot

For each robot controller that requires the J1 Backlash Test, perform [Section 3.3.6.1.1, J1 Backlash Test](#).

3.3.6.1.1 J1 Backlash Test

About this task

Use this procedure to test J1 backlash.

Procedure

1. Rename the following job programs on the controller:

g1_j1_new_job.tp

g2_j1_new_job.tp

Note

The J1 test job should be job 99. If job 99 is already in use by the plant, the job number can be changed (just remember to set the var: `sj1JobNum` and in the PAVRZDT variable file in the controller). Rename the job programs P1990000 and P2990000 to have names consistent with site conventions for the P-700 robot groups 1 and 2 on the controller. For instance, the "P" may need to become a "J" or a "G" depending on the site and process conventions. The "1" may need to become a "3", "5", or "7" depending on the name of the corresponding robot for group 1. Similarly, for group 2 the "2" may need to become a "4", "6", or "8".

Note

The J1 test jobs are for painters only (not openers). Make sure that the controller does not have a job 99 assigned for an opener group. Also make sure the Paintworks GUI does not have openers enabled for either of the test jobs.

2. Rename the two lines in the J1 job program, CALL HOME P1/P2, to the appropriate home program if site conventions have a different home program name for each P-700 robot.

Similar to the job name, the HOME P1 may become ...P3, ...P5, or ...P7, and HOME P2 may become...P4, ...P6, or ...P8.

3. The R099 customization loads two TP programs named BACKLHG1_ZDT and BACKLHG2_ZDT. If the controller already has the J1 test program loaded (look for BACKLHG1 and BACKLHG2), these two new programs can be deleted. If not, rename the two newly loaded programs by removing the “_ZDT” part of the name.

Note

For V7.30 software, BACKLHGx_ZDT is named BCKLHGx and Gx_J1_NEW_JOB is named GxJ1_JOB to adhere to the strict 8 character long file naming convention.

4. Check and/or touchup the two points in BACKLHG1 and BACKLHG2 which are teach pendant processes for P-700 group 1 and group 2.

Make sure the positions remain in joint representation. These positions must move only joint 1 of the P-700 to nearly the full range of joint 1 motion but staying slightly away from DCS limits and axis limits. The joints 2-7 must be configured to avoid interference with other robots and obstacles in the workcell. Make sure there will be no interference with other robots running the programs at the same time. Make sure there will be no issue with running the HOME1/P2 or (...P3 – ...P8) programs to/from P[1] taught in BACKLHG1/G2.

5. Run the Job 99 J1 Backlash Test from the Paintworks GUI.

The test moves the robot to the first taught position of J1, then moves J1 a degree at a time until the last taught point is reached. After the job is complete, the Paintworks system runs a checking program and updates the status of the test on the ZDT alert screen. The J1 test data is also copied to a log file.

3.3.7 Paint Canister Torque Health Monitoring Systems

PaintWorks Required

FANUC waterborne paint canisters are designed for high reliability. Due to environmental conditions, the mechanical integrity of the canister may be compromised over time. ZDT monitors the torque values during operation of the canister and will alert you of higher trending torque data well ahead of incurring a fault which will interrupt the paint process, causing downtime and potential quality issues.

ZDT will alert you of canister issues in advance so maintenance can be scheduled before incurring production downtime. The cost of a canister torque fault in production can be substantial.

3.3.8 Application Status

This feature allows ZDT to show the summary of the operational efficiency of the robot, to show how the robot has been running for the previous hours. [Table 3.3.8](#) shows the items used to configure the Application Status.

Table 3.3.8 Application Status Items

Item	Description
In-cycle measured by	This item determines how the program “In-Cycle” time statistics are measured in “Production Data”, which is in “Process”. The choices are: • BUSY– Robot is deemed to be in-cycle when program running. • MOTION– Robot is deemed to be in-cycle when moving.

Item	Description
Cycle Count by	This item determines how the robot cycle count is measured. The choices are: - PROGRAM START – Every time a program is run in AUTO, the cycle counter is incremented. If the program does not lock any group, it will not be counted as a cycle. If a program uses RUN_TASK, other method should be used. - UOP PROD_START – Every time a program is run in AUTO mode by the Prod_start UOP input, the cycle counter is incremented. - INPUT PORT – While in AUTO mode, the cycle counter is incremented when a pulse is input on a digital input port. - USER PROGRAM – Use a register to store the cycle count. The user program is responsible for updating the value.
Cycle count input	This item is only used when the previous item is set to “INPUT PORT”, and indicates the index number for the input port to monitor for a pulse.
Register Index	This item is only used when the previous item is set to “USER PROGRAM”, and indicates the index number for the register for the cycle counter.
Current cycle count	This indicates the accumulated cycle count value. The operator may modify this value here in this screen.

3.3.9 Vision Find Time Monitor

High find times in a vision process can be a sign of a multitude of problems, such as an unfocused lens, busy background, or changed parts. If find time becomes too high, fewer parts will be found over a given time, resulting in loss of productivity. ZDT monitors vision processes and alerts you if a high find time condition exists or is predicted to occur soon by comparing sample averages to a threshold. This threshold is calculated from the vision data.

ZDT will alert you of high find times so that the vision system can be investigated and the proper maintenance can be performed.

A small amount of setup via the teach pendant is required. Refer to [Table 2.1.2](#).

3.3.10 EOAT Dropped Parts Count

Slow gripper actuation time, low vacuum pressure, worn vacuum cups, or other common issues with end-of-arm tools can cause parts to be missed or dropped, especially in high speed picking applications. Using flex data, ZDT can track the amount of dropped parts and alerts you when an excess occurs. Dropped part counts are determined on daily, hourly, and cumulative bases and compared to threshold values, as specified by the user.

ZDT will alert you when too many parts have been dropped so the EOAT can be checked and repaired if necessary.

Setup via the teach pendant and the ZDT portal is required.

3.3.10.1 Controller Setup

To collect part count data, three items are required:

1. A sensor to detect the presence/absence of parts

2. Teach pendant logic to write values to registers
3. Flex data configuration to send register contents to ZDT

A sensor (part presence or similar) must be used as input to the controller to determine the presence or absence of parts. Teach pendant logic must be written to check this input and increment a register. Two registers will be used – a register for tracking dropped part counts and a register for tracking total part or cycle counts. For every cycle, the total part/cycle count should be incremented, and logic should determine whether or not to increase the dropped part count. Logic to prevent overflow in the registers is recommended.

In the following example, the value of R[5] contains dropped part data, and is incremented when the part presence sensor (DI[1]) is OFF. Regardless of the sensor state, the value of R[6], containing the total part/cycle count, is incremented. Both registers are first checked for overflow.

```
4: IF (DI[1]=OFF) THEN
5:
6: IF (R[5]>=10000) THEN
7: R[5]=0
8: R[5]=R[5]+1
9: ELSE
10: R[5]=R[5]+1
11:
12: ENDIF
13: ENDIF
14:
15: IF (R[6]>=10000) THEN
16: R[6]=0
17: R[6]=R[6]+1
18: ELSE
19: R[6]=R[6]+1
20:
21: ENDIF
```

Figure 3.3.10.1 Dropped Parts Count example

Once TP logic is written to increment these two registers, flex data must be configured to send their contents. Please view the instructions for configuring flex data and flex I/O in [Section 3.1.4, Flexible Data](#).

3.3.10.2 Portal Setup

After configuring the robot to send dropped part and total part/cycle counts via flex data, the robot must be configured in the portal. See [Section 3.3.10.3, Configuring the Robot in the Portal for EOAT Dropped Parts Count](#).

3.3.10.3 Configuring the Robot in the Portal for EOAT Dropped Parts Count

About this task

Procedure

1. Navigate to the **Robot Info** page on the portal.
2. Find the robot you wish to configure for EOAT Dropped Part Monitor.
3. Click **Toggle Grid Editing**.

4. In the **Process/Special Characteristics** column, choose **EOAT_Part_Drop**.

```
Vision ; Sends_Alarm_Msg_T ; Sends_Power_T ;
Sends_GJOB_DATA_T ; Sends_batt_low2_t ;
EOAT_Part_Drop ; Sends_Bins_or_Job ;
Sends_rtc_t
```

Figure 3.3.10.3 (a) EOAT Part Drop

5. A robot will become configured for this analytic after it has been configured in the **Robot Info** page. Once available, a link to the EOAT Dropped Part Monitor data view will be available for this robot under the **Process Health** category. Follow this link to the data view.
6. If this robot has not yet been configured in the data view, you will need to provide it the registers you are using for dropped part and total part/cycle counts, as well as desired thresholds for daily, hourly, and running total analysis.
7. Click on **Configure Analytic**.
8. Enter the registers and thresholds you will be using and press **Apply Settings**.
9. If this robot has not yet been configured in the data view, you will need to enter values for all five settings.

If the robot has been configured, then you can update settings individually.

Dropped Parts Register New: R [] Current: 5 Apply Settings	Total Parts/Cycles Register New: R [] Current: 6	Daily Threshold New: Current: 20	Hourly Threshold New: Current: 10	Running Total Threshold New: Current: 100
---	---	--	---	---

Figure 3.3.10.3 (b) EOAT Settings

3.4 Maintenance Health

3.4.1 Maintenance Reminder

The Maintenance Reminder feature alerts users that periodic or intelligent robot or system maintenance is due. The majority of robot models have predefined items set up for recommended maintenance tasks (Required Maintenance).

You can customize the Maintenance Reminder for non-robot items, such as an actuator or valve, you would like to have a reminder on when it is time for maintenance.

Please refer to any *Setup and Operations* manual and locate the *Maintenance Reminder chapter*. The documentation may indicate you need to install the software option J771, however, this is not necessary with ZDT. Maintenance Reminder included with ZDT is a subset of the full J771 Maintenance Reminder option. When J771 is not installed you will not receive local alerts on the robot controller or the ability to view.

To enable Maintenance Reminder, please visit the **SETUP ZDT Client** screen described in [Section 2.1, ZDT Client Setup on the TP](#).

4 Internal Errors

In this chapter, internal errors are defined.

Table 4 Description of Errors

Error	Description	
ZDTURL parse error%d	This is a general error and it carries specific code to identify the error in URL.	
	100	Method is not valid
	101	URL does not have <code>http://</code>
	102	URL does not format <code>http://.../</code>
	103	Host name too long
	104	No host name found
	105	XML can not be parsed. Make sure host name resolvable by DNS.
ZDT Web Server Error %d	106	HTTP Queue full
	282	Timeout: Data Collector or Web Server did not respond to PING request robot send
	Others	All other errors are standard error. All standard errors are defined in error manual and HTTP status codes are listed in RFC2626.

PART II

DATA COLLECTOR SETUP

Topics:

- [Get Started](#)
- [Using the Dashboard](#)
- [Data Collector](#)
- [User Management](#)
- [Administration Tasks](#)
- [Robot Configuration](#)
- [Commands](#)
- [Troubleshooting](#)

The Data Collector provides real-time data regarding the performance and state of the connected robots, allowing you to easily monitor information about the devices in your deployment. This section discusses setup related to the Data Collector.

1 Get Started

The FANUC Zero Down Time (ZDT) Data Collector provides real-time data regarding the performance and state of the connected robots, allowing you to easily monitor information about the devices in your deployment.

Before you can log in to the ZDT Data Collector, you must obtain the information and files necessary to complete the network and system configuration. Prepare your network to support the ZDT Data Collector, and install the virtual machine (VM).

After these requirements are complete, you can log into the ZDT Data Collector to change the default password, configure the network interface, and install the configuration file and security certificate. You can also enable an LDAP server and create additional users, and install software updates, if necessary.

Related concepts

[Initial Setup](#)

[Backup and Restore the Satellite Configuration](#)

[Getting Help](#)

Related reference

[Requirements](#)

1.1 Requirements

The Zero Down Time (ZDT) Data Collector is a web-based UI that requires the following.

Table 1.1 Requirements

Requirement	Description
Supported browser	(Required) Latest version of Chrome or IE 11. Note If you receive a message that a log or other file cannot be downloaded when using IE 11, install the latest Microsoft security update.
Supported resolution	(Required) Minimum 1024x768
A VMWare ESXi v 6.5 or 6.7 server appliance for installing the virtual machine (VM)	(Required) The VM is deployed using the VMWare ESXi web interface, and requires the following: • VMWare ESXi compatible web browser • 6 CPUs • 8 GB RAM • 60 GB disk space See Section 1.2.4, Install the Virtual Machine .
OVA file used to install the VM	(Required) See Section 1.2.4, Install the Virtual Machine .
ZXL Site info file (.zxl)	(Required) See Section 1.2.2, Obtain the Required Site Information and Section 1.2.7, Uploading the ZDT Provisioning File .

Requirement	Description
SSL/TLS certificate for HTTPS authentication	(Optional) See Section 1.2.2, Obtain the Required Site Information and Section 1.2.8, (Optional) Install a SSL TLS Certificate for HTTPS Authentication .
Network information and access	• (Required) TCP port access • (Required) IP address and other NIC settings • (Required) NTP server address • (Optional) LDAP server address See Section 1.2, Initial Setup .

1.2 Initial Setup

Complete the following procedures to configure your network and ZDT Data Collector. These basic steps must be completed before any ZDT messages can be collected and sent to the FANUC data center for analysis and support.

1.2.1 Summary Steps

About this task

Step summary

Procedure

1. Obtain the Required Site Information.
2. Prepare the Site Network.
3. Install the Virtual Machine (VM).
4. Log in and Change the Default Password.
5. Configure the Network Interface.
6. Upload the ZXL Site Info (ZDT Provisioning) File.
7. Install a SSL/TLS Certificate for HTTPS Authentication.
8. Update Data Collector Software.
9. (Optional) Enable LDAP and Add Users.
10. (Optional) Disable the Admin User.

1.2.2 Obtain the Required Site Information

Gather the following information that is unique to your site and deployment. This information is used to install and configure the ZDT Data Collector.

Table 1.2.2 Required Site Information

Required Information	Note Entry
Information required by FANUC to provide the ZXL Site Info (ZDT provisioning) file (.zxl): • Customer name • Country • Plant name	
Network configuration • Static IP address • Host name • Subnet mask • Gateway IP address • DNS server IP address(es) • DNS search extension(s) • Proxy server URL, if required	
(Optional) Outbound traffic IP address Note One NIC can be used for both inbound (listener) and outbound traffic. In this case, the outbound IP address is not required. To send outbound traffic on a separate NIC, configure the NIC using DHCP or a static IP address. Static IP addresses require the following information: • Static IP address • Host name • Subnet mask • Gateway IP address • DNS server IP address(es) • DNS search extensions(s)	
NTP server IP address	
(Optional) SSL/TLS certificate for HTTPS authentication	
(Optional) LDAP Information is required to configure additional user login accounts. • Server name(s) and port number(s) • Search account DN • Search account password • Search base DN • User object class • User ID attribute • CER formatted CA Certificate for the LDAP server	

1.2.3 Prepare the Site Network

The following TCP network ports must be open to allow the ZDT Data Collector to communicate with the robots and the cloud. This network configuration also provides secure web access to the browser-based ZDT console.

Table 1.2.3 Required TCP Network Ports

TCP Network Ports	Purpose
6062, 6063 and 5672	Robots send ZDT messages to ports 6062, 6063 and 5672 of the data collector.
3002	The data collector sends and receives data from robots using port 3002.
443	- The data collector sends secure messages to the cloud using URL <code>fanuc01.cisco.com</code> or <code>zdtdeprimary.fanuczdt.com</code> on port 443. - The console UI is accessed with HTTPS on port 443 on the data collector.
21	Data sent and received via FTP.
LDAP port (typically 636)	(Optional) If LDAP is enabled in ZDT to allow additional users to log in to the console, the LDAP server port (typically 636) must be accessible from the ZDT Data Collector.

1.2.4 Install the Virtual Machine

The ZDT Data Collector runs as a virtual machine (VM) which is installed as an OVA file on a VMWare ESXi server appliance.

VM Reuirements

The OVA is deployed using the VMWare ESXi web interface, and requires the following:

- 6 CPUs
- 8 GB RAM
- 60 GM disk space

Versions

A base Virtual Machine (VM) for each major release. Minor release at the zero update level will be provided by FANUC. Versions of the ZDT Data Collector are numbered X.Y.Z:

X = Major release number

Y = Minor release number

Z = Update level

1.2.4.1 Installing the Virtual Machine

About this task

Use this procedure to install the virtual machine.

Procedure

1. Open web browser and point it to your VMWare ESXi server on your PC or laptop.
2. Click **Virtual Machines > Create/Register VM**. On the dialog box, select **Deploy a virtual machine from an OVF or OVA file**. Click **Next**.

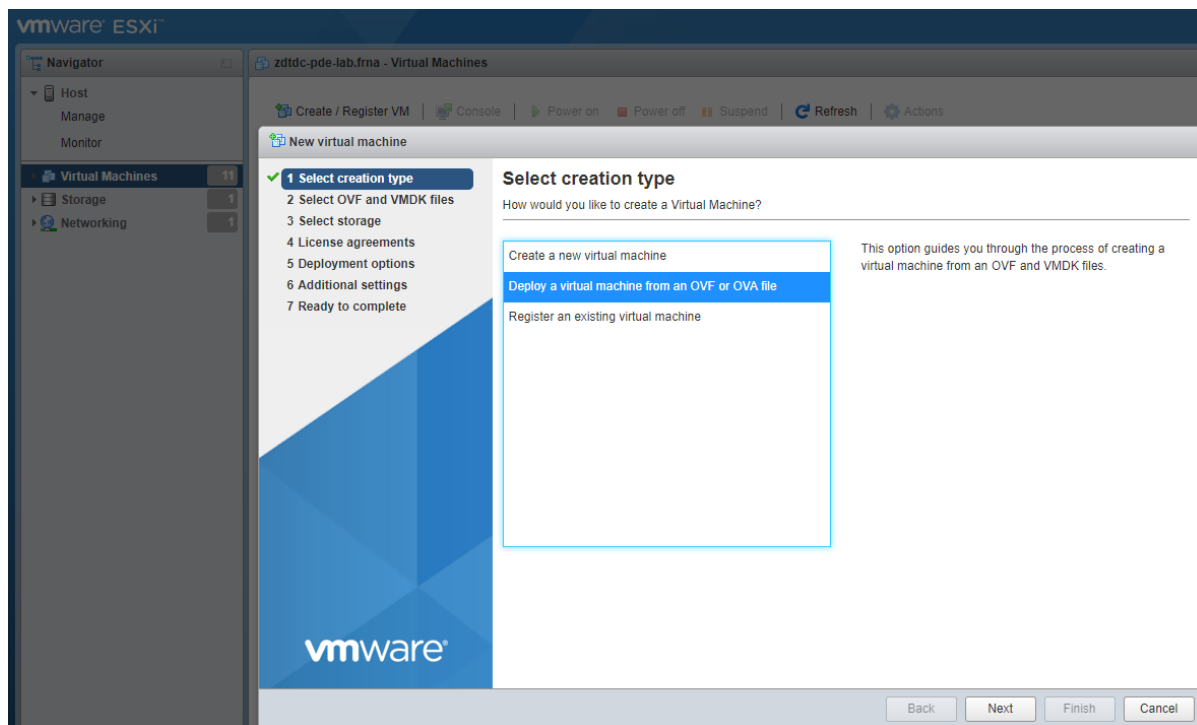


Figure 1.2.4.1 (a) VMWare ESXi

3. On **Select OVF and VMDK files** dialog box, enter a name for the virtual machine and select the Data Collector OVA file, click **Next**.

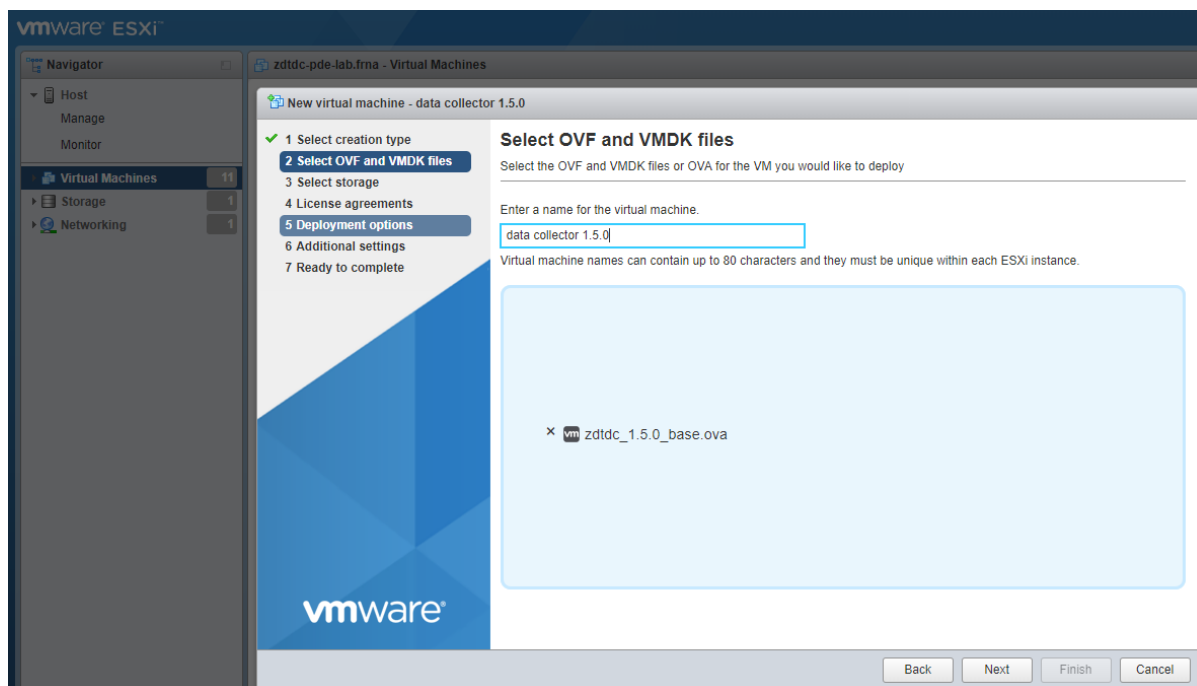
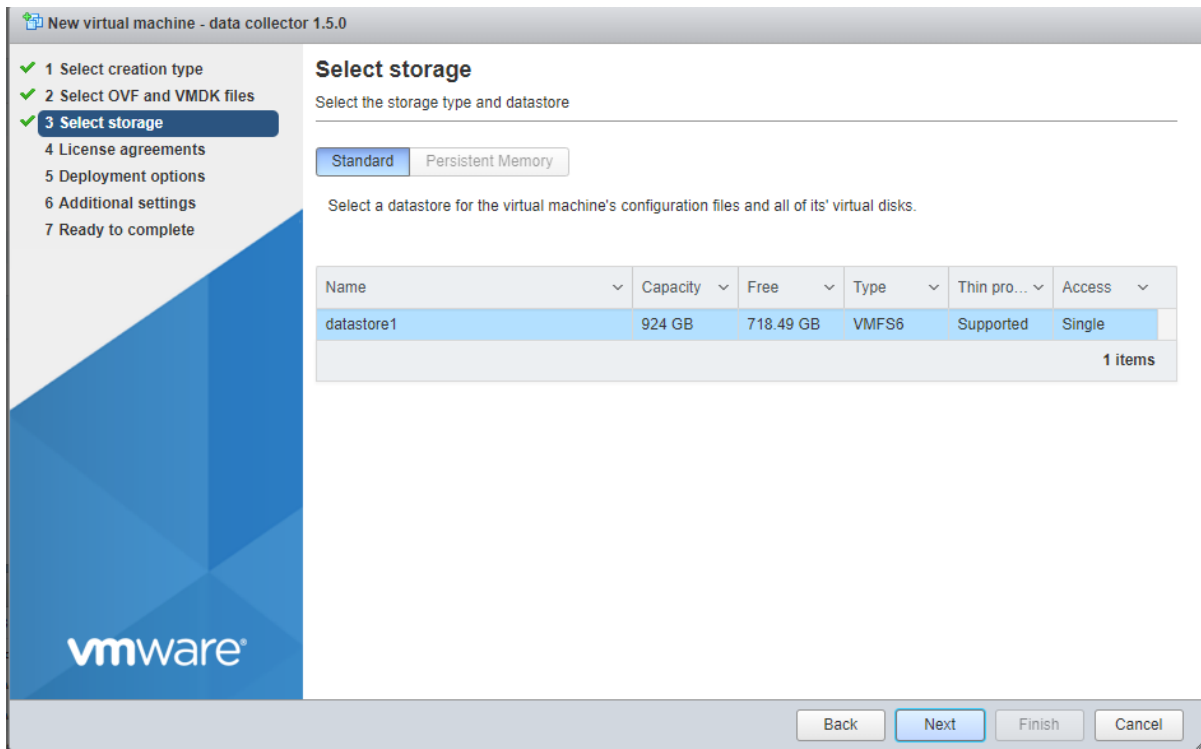
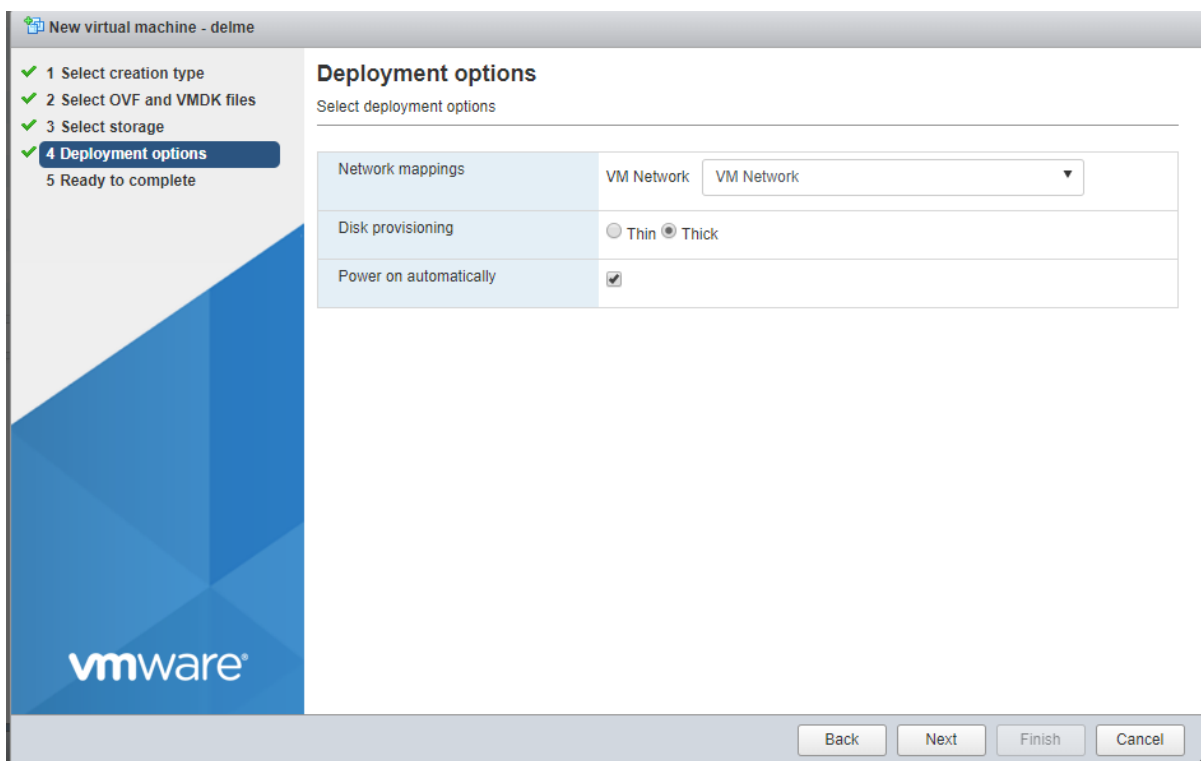


Figure 1.2.4.1 (b) Select OVF and VMDK files

4. Select storage, click Next.**Figure 1.2.4.1 (c) Select Storage****5. Select Deployment options. We recommend Thick provisioning. Click Next.****Figure 1.2.4.1 (d) Deployment options****6. Review your settings and click Finish.**

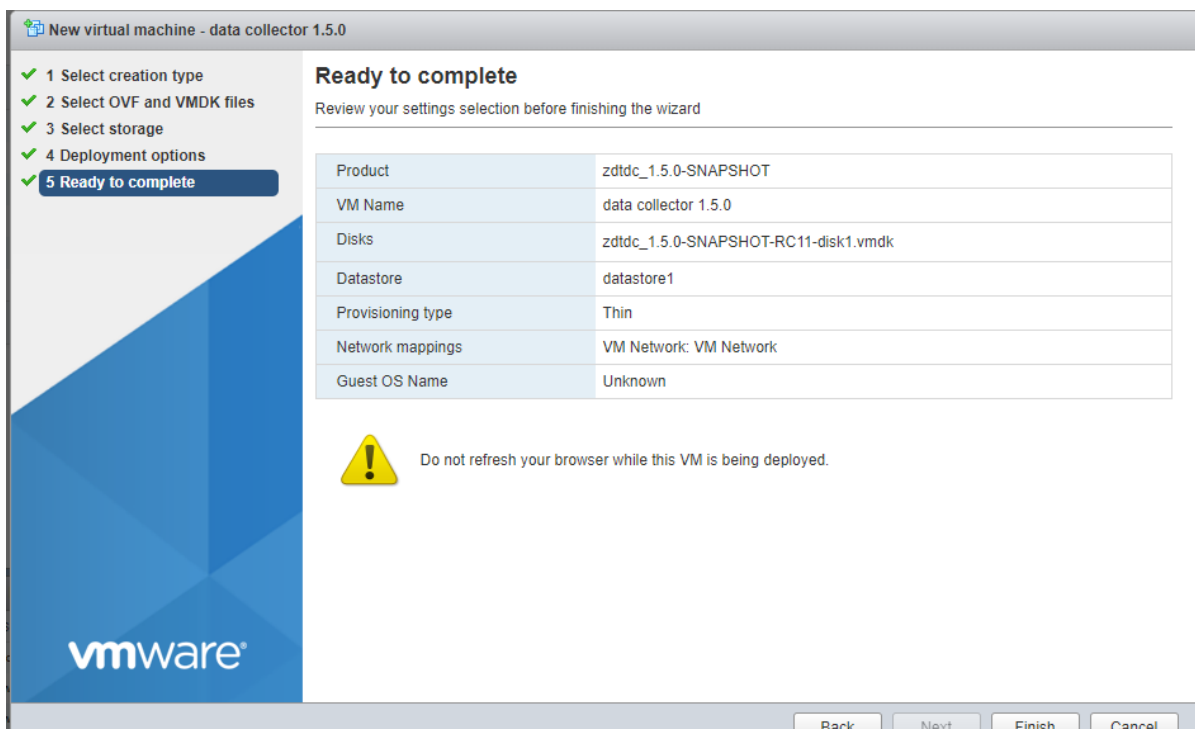


Figure 1.2.4.1 (e) Ready to complete

7. Select and power on the Data Collector VM, click **Console**. Continue to configure a static IP address or DHCP for the VM.

Note

You must select and complete one of these options. The network settings can be changed later, but failure to set the IP address using this screen can potentially allow an unauthorized person to change the network settings.

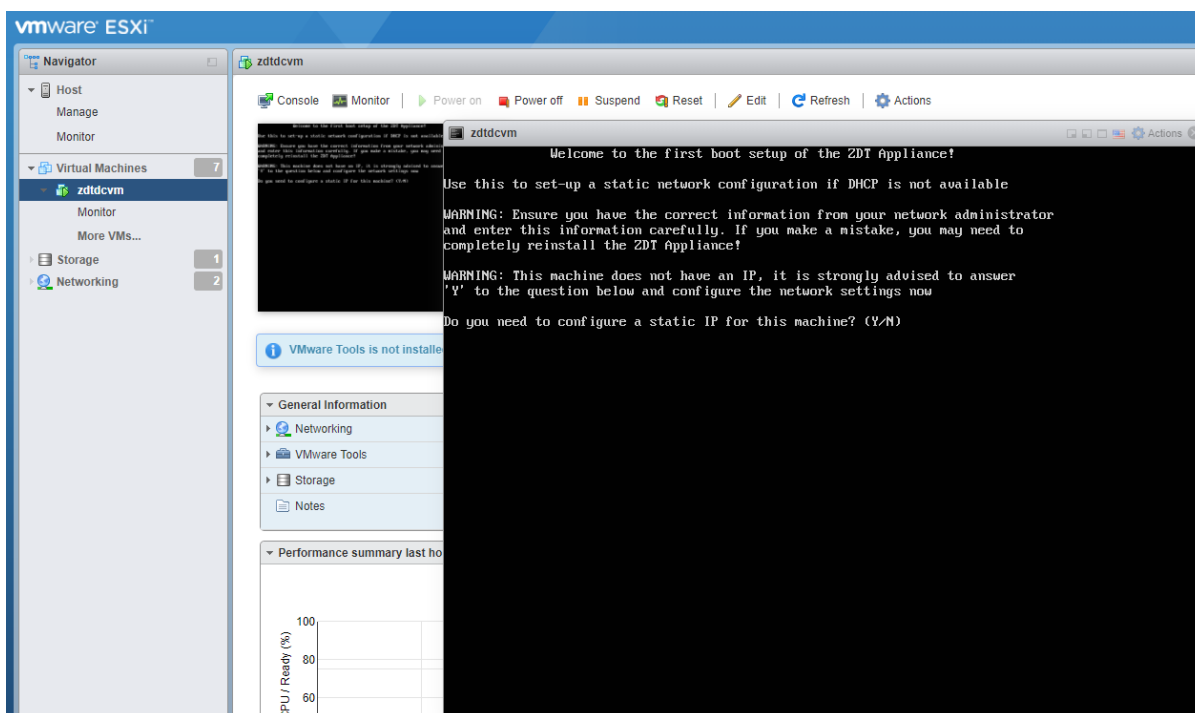


Figure 1.2.4.1 (f) Configure the IP Address

8. Change the data collector's `zdtuser` default password. From the console, log into the data collector using the username `zdtuser`. The default password will be provided by FANUC with the OVA file.

You will be prompted to change the `zdtuser` password – Follow the command line prompts to complete this step.

Note

The system has strict password guidelines. The new password must contain uppercase, number and symbol.

1.2.5 Log in and Change the Default Data Collector Web UI Password

Log in to the ZDT Data Collector and change the default admin password.

1.2.5.1 Logging In and Changing the Default Password

About this task

Use this procedure to log in and change the default password.

Procedure

1. Launch a web browser and enter the IP address created when you installed the VM (see [Section 1.2.4, Install the Virtual Machine](#)).
2. Log in using the default administrator password.
Username: `admin`
Password: `admin`
3. Change the Admin Password.
 - a) Click **Admin**.
 - b) Enter the current admin password.
 - c) Enter and re-enter a new password.
 - d) Click **Change Password**.

Note

(Optional) You can also Enable LDAP and Add Users and (Optional) Disable the Admin User.

1.2.6 (Optional) Configure the Network Interface

If you need to make changes to the network settings use the following procedure.

1.2.6.1 Configuring the Network Interface

About this task

Use this procedure to configure the network interface.

Procedure

1. Click **Network Configuration**.
2. (Optional) Select **Use Different Inbound/Outbound Addresses** to use different IP address for outgoing traffic.
3. Enter the network settings for your deployment, including:
 - Hostname—the portal hostname
 - Static IP address—the static IP address for the portal. DHCP is not allowed for the inbound NIC.
 - Subnet mask
 - Gateway IP address
 - DNS server IP address(es)
 - DNS search extension(s)
 - NTP Server address(es)
4. (Optional) Select or deselect **Use Different Inbound/Outbound Addresses** and enter the network settings for outbound traffic. DHCP can be used to provide the network settings.
5. Click **Save**.
6. (Optional) Enter a proxy server URL.

Select this option to use a proxy server to send traffic from the satellite to the primary server.

 - a) Select **Use an HTTPS proxy URL**.
 - b) Enter the server URL or IP address.

The proxy must be either the HTTP or HTTPS protocol.

A port number is required. For example: <https://example.com:443>.
 - c) Click **Save** and then **OK** to confirm.

Use this option to send data to a local proxy server before it is sent to the cloud. This is a security requirement in some environments.

What to do next

Continue to [Section 1.2.7, Uploading the ZDT Provisioning File](#).

1.2.7 Uploading the ZDT Provisioning File

About this task

The ZDT provisioning file is a .zxl site configuration file provided to you by FANUC that contains basic information about your organization and deployment. This file must be uploaded to the ZDT Data Collector for proper operation.

Procedure

1. Click **ZDT Data Collector**.
2. Click the **Site Info** tab.
3. Click **Upload ZDT Provisioning File**.

4. Select the file and click **Upload**.

What to do next

Continue to [Section 1.2.8, \(Optional\) Install a SSL TLS Certificate for HTTPS Authentication](#).

1.2.8 (Optional) Install a SSL TLS Certificate for HTTPS Authentication

Upload the security certificate that authenticates the Satellite REST API server and allows secure communication when browsing to the web console UI and accessing the API.

1.2.8.1 Uploading a Security Certificate

About this task

Use this procedure to upload a Security Certificate.

Procedure

1. Obtain the *SSL/TLS certificates* from your IT department or purchase it from SSL/TLS vendors.
2. Click the **Certificates** tab.
3. Click the **API** tab.
4. Click **Replace HTTPS Server Certificate**.
5. Click **Browse** and select the file from a local drive.
6. Click **Upload** and then **OK**.

What to do next

Continue to [Section 1.2.9, Update Data Collector Software](#).

1.2.9 Update Data Collector Software

This step is only required if

- You receive the Data Collector OVA with updater file for initial installation.

Or

- Your support representative send you the Data Collector updater file to update Data Collector.

The ZDT Data Collector can be updated for UI and functional changes using a signed image provided by your support representative. The update file must be a valid format that includes the correct security key. See [Section 5.3.1, Updating the System Software](#) for more information.

1.2.9.1 Updating the Portal Software

About this task

Use this procedure to update the Portal software.

Procedure

1. Click **Admin** and then **Update**.
2. Select the update file from a local or network drive.
3. Click **Upload**.
4. Click **OK** when the upload is complete.

What to do next

Continue to [Section 1.2.10, \(Optional\) Enable LDAP and Add Users](#).

1.2.10 (Optional) Enable LDAP and Add Users

An LDAP server allows you to add additional users using their LDAP user name. Users can be assigned a variety of roles and access permissions.

1.2.10.1 Configuring an LDAP Server to Add Users

About this task

Use this procedure to Configure an LDAP server to add users.

Procedure

1. Configure an LDAP server to provide access to additional users.
 - a) Click **User Management**.
 - b) Click **LDAP Configuration**.
 - c) Complete [Section 4.2, Configure the LDAP Server Connection](#).
2. Add additional users.
 - a) Click **User Management**.
 - b) Click **User Roles**.
 - c) Click **Add User** and enter an existing LDAP username.
 - d) Select one or more user roles. See [Section 4.4, User Roles](#), for more information.
 - e) Click **Add User**.
 - f) Add additional users as necessary.
 - g) When done, click **Cancel** to close the **Add New User** window.

1.2.11 Backup and Restore the Satellite Configuration

We highly recommend backing up the satellite's configuration before and after you make a major configuration change. This allows you to roll back to a previous working configuration if problems occur, or re-apply the configuration to a different Satellite OVA instance.

For example, backup the configuration after the Initial Set Up, when you modify robots, add users to the user role list, setup an LDAP server, change the security certificates, change the primary, change the proxy, change the network configuration, or change the time zone.

If issues occur, you will have a known and stable configuration to restore.

1.2.11.1 Usage Notes

- The backup archive must come from a satellite with the same version as the recovery satellite. The only exception is restoring backup from 1.4.8 to 1.5.0. If the versions differ, the restore will fail.
- If the backup archive is from a satellite that has a different hostname or IP address, a warning message appears. You can cancel or click **OK** to continue. If you continue with a mismatched address.
 - Reload the ZDT web console UI.
 - If the restore changed the hostname/IP that you used to access the UI from your browser, you will need to use the new hostname/IP to access it after the restore. If necessary, you can go to the vSphere console (see [Section 1.2.4.1, Installing the Virtual Machine](#) step 7), and the IP address will be displayed.

1.2.11.2 Backup the satellite configuration

The backup archive contains the satellite's network, HTTPS proxy, and LDAP configurations, primary address, time zone, user roles, all certificates and robot operations.

1.2.11.2.1 Backing Up the Satellite Configuration

About this task

Use this procedure to backup the Satellite Configuration.

Before you begin

Data collector's network configuration is complete.

Procedure

1. Click **Admin > Backup**.
2. Click **OK** when the backup is complete.
3. Click **Admin**.
4. Create a new user or modify an existing user to include the System Administrator user role.
See [Section 4.3, Manage User Access](#), for more information.

1.2.11.3 Restoring the Backup Archive

About this task

Restore the backup archive to apply the backup configuration to your satellite.

Before you begin

Satellite configuration has been previously backed up.

Procedure

1. Click **Admin > Restore**.
2. Select a backup file from a local drive and click **Restore**.

1.3 Getting Help

For help and more information, contact support: 1-800-477-6268 press **1** for Robot.

2 Using the Dashboard

The Dashboard provides a snapshot of the ZDT Data Collector system performance and communication with the associated robots.

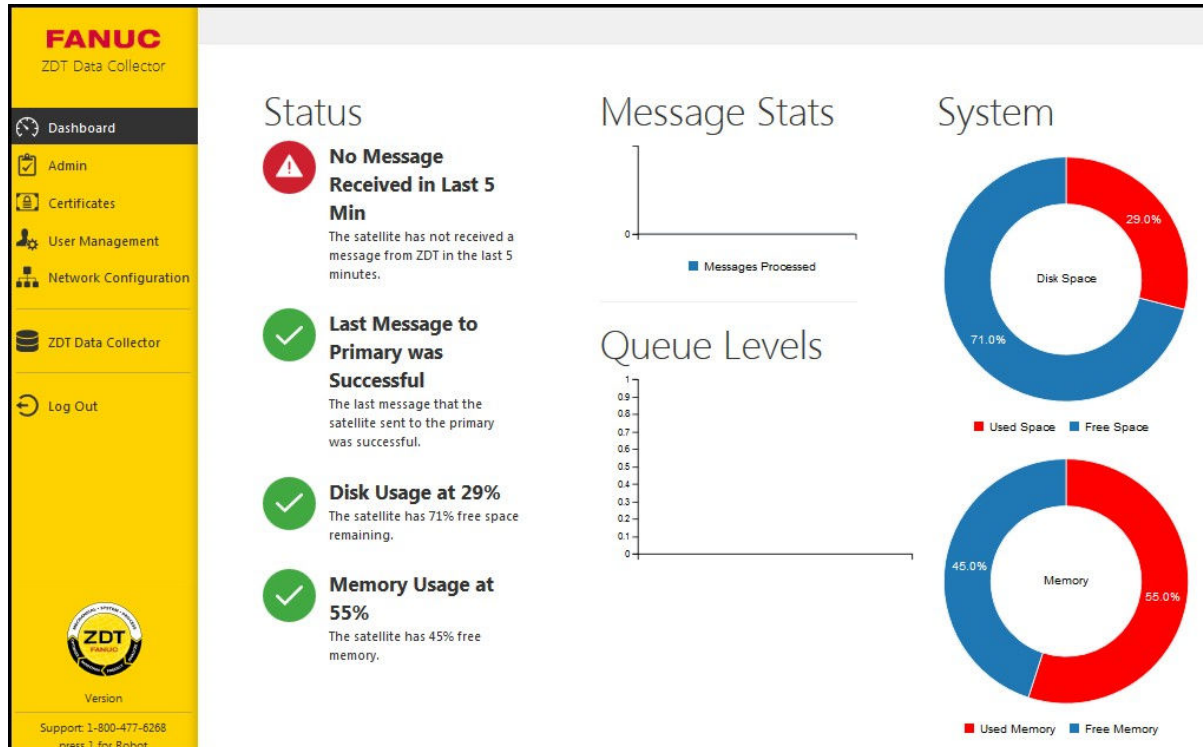


Figure 2 Dashboard

2.1 Log In

Log in to the Zero Down Time (ZDT) Data Collector using the username and password provided by your system administrator.

2.1.1 Logging in to the Data Collector

About this task

Use this procedure to log in to the Data Collector.

Before you begin

The username and password must be known.

Procedure

1. Launch a Chrome or IE web browser and go to the URL provided by your administrator.
2. Enter your username and password.

2.2 Status

The Status field provides a high-level snapshot of critical tasks and system performance

Table 2.2 Status

Status	Description
Message received	• Green—Messages have been received from robots within the past 5 minutes. • Red—No messages have been received from robots within the past 5 minutes.
Last Message Sent	• Green—The last message sent to the primary server succeeded. Communication is successful. Green is also displayed if no messages were previously sent. • Red—The last message to the primary server failed.
Disk Usage	• Green—Disk usage is under 90%. • Red—Disc usage is over 90%.
Memory Usage	• Green—Memory usage is under 90%. • Red—Memory usage is over 90%.

2.3 Message Stats and Queue Levels

Messages are sent by the robots to the ZDT Data Collector. These fields show the volume of messages received and the rate that they are being processed.

- Message Stats—The number of messages received from the robots over time.
- Queue Levels—The percentage full for each message queue (0-100%). For example, the message queue displays a large volume of messages if the ZDT Data Collector is not processing the influx of data fast enough. For example, if a robot, or group of robots, is sending a large volume of data.

2.4 System

The charts in the **System** field show the memory and disk usage data as a graph.

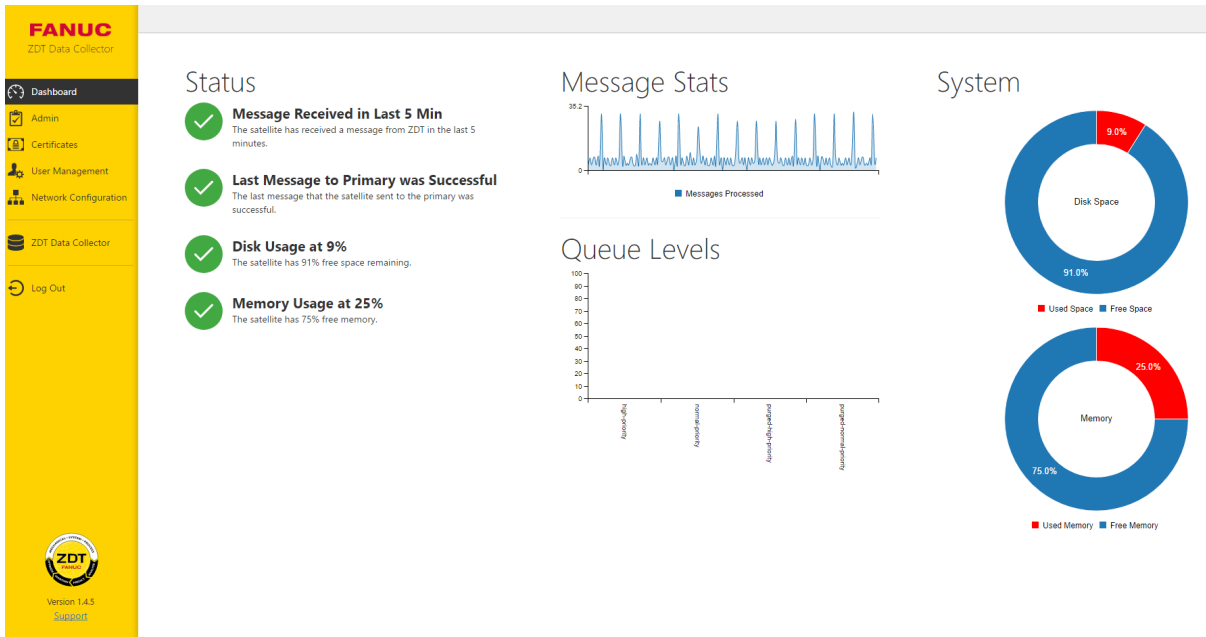


Figure 2.4 System charts

3 Data Collector

The ZDT Data Collector provides dynamically updated information about the system, and the robots that provide data. Click the column headers in each section to sort the data by that category.

3.1 ZDT Data Collector Process Statuses

The **Process Status** page displays the processes running on the ZDT Data Collector server. Use this page to troubleshoot system performance or other system issues.

Note

Click **Restart Processes** to restart the running processes.

The Service entries 1-6 display the following information:

Table 3.1 (a) Process Status

Data	Description
Service	Service name
Process ID	The process ID
% CPU	The percentage of system CPU that the process is using
% Memory	The portion of system memory that the process is using
Memory Size	The amount of memory that this process is using
Start Time	When the process began running
Elapsed Time	How long it has been running

The Queue entries 7-12 display the following information:

Table 3.1 (b) Queue Status

Data	Description
Queue	Message queue name
Message Count	Number of messages in this queue
Rate In	Incoming message rate
Rate Out	Outgoing message rate
Memory Size	Amount of memory used by this queue

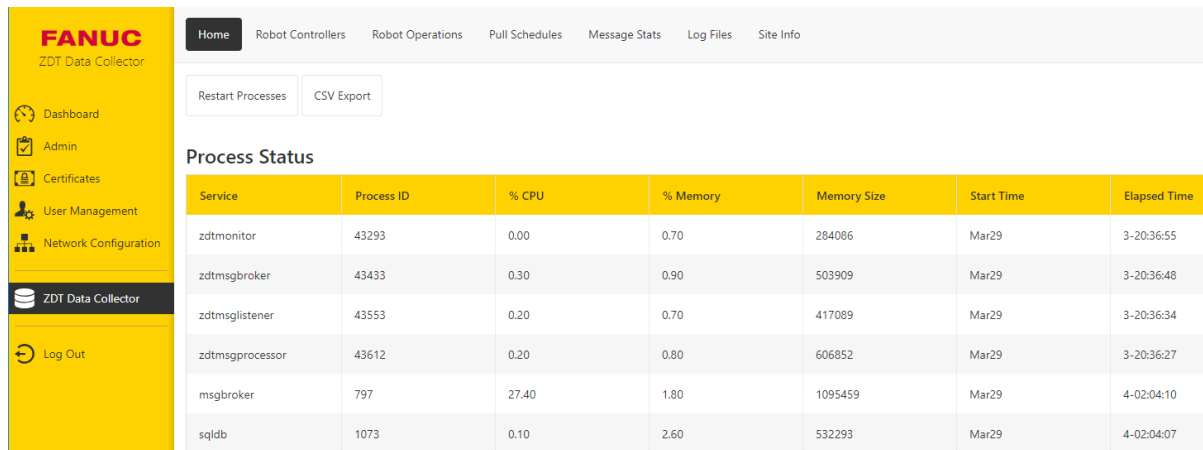


Figure 3.1 ZDT Data Collector Process Statuses

3.2 Robot Controllers

This page includes all robots that are currently sending data to the system. The data is dynamically updated every 5 seconds.

Table 3.2 Robot Controllers Page Descriptions

Name	The robot name
Name	The robot name
Controller IP	The robot IP address
Controller ID	The globally unique ID of the robot
Controller Type	The device type. Legacy = R-30iA, ZDT = R-30iB
Controller State	The current state of the device, such as Enabled or Disabled
Application	The name of the application tool installed on the robot. For example; SpotTool+
Version	The system software version installed on the robot
Last Updated	When last message received from the robot

Tip

Enter a phrase or partial phrase in the Filter to display only robots that include that information. The filter applies to all fields. For example, enter a partial IP address to include devices with that root address.

3.3 Robot Operations

Use the Operations page to manage robot controllers .



FANUC
ZDT Data Collector

- Dashboard
- Admin
- Certificates
- User Management
- Network Configuration
- ZDT Data Collector**

Home Robot Controllers **Robot Operations** Pull Schedules Message Stats Log Files Site Info

Manage Robot Controllers

1.1

1_IP,Job_Data,Condon,Data_Frequency_Type,Frequency_Number,D,H,M,MRA,Line,Cluster
1715:00-1846:00-1715:00-1846:00-1715:00-1846:00

Export

Submit

Most Recent Robot Operations

Started By	Controller IP	Operation	State	Result
admin	1715:00-1846:00	Add	Done	Success
admin	1715:00-1846:00	Add	Pending	Not Started
admin	1715:00-1846:00	Add	Pending	Not Started
admin	1715:00-1846:00	Delete	Done	Failed
admin	1715:00-1714	Delete	Done	Failed
admin	1715:00-1846:00	Delete	Done	Failed
admin	1715:00-1846:00	Delete	Done	Failed
admin	1715:00-1714	Delete	Done	Failed
admin	1715:00-1846:00	Delete	Done	Failed
admin	1715:00-1846:00	Delete	Done	Failed
admin	1715:00-1846:00	Delete	Done	Failed
admin	1715:00-1846:00	Delete	Done	Failed
admin	1715:00-1846:00	Delete	Done	Failed
admin	1715:00-1846:00	Delete	Done	Failed
admin	1715:00-1846:00	Delete	Done	Failed
admin	1715:00-1846:00	Delete	Done	Failed

Copy and paste DiagnosticPRO server configuration content into this text field to preserve robot controller IDs.

```

INFO 2016-02-12 16:54:49,718 2 robot configuration records read.
WARN 2016-02-12 16:54:49,773  already defined so preserving its existing robot controller ID 1715:00-1846:00
WARN 2016-02-12 16:54:49,774  already defined so preserving its existing robot controller ID 1715:00-1846:00
INFO 2016-02-12 16:54:49,774 Processing requests.
INFO 2016-02-12 16:54:49,775 Checking if robot controller exists. ip: 1715:00-1846:00
INFO 2016-02-12 16:54:49,795 Checking if robot controller exists. ip: 1715:00-1846:00
INFO 2016-02-12 16:54:49,804 Checking if any robot within the request has an existing pending operation.
FATAL 2016-02-12 16:54:49,826 Robot  has an existing operation that is pending. Please re-submit after the operation has finished.

```



Version 1.4.1
[Support](#)

Figure 3.3 Robot Operations

3.3.1 Manage Robot Controllers

Enter configuration commands the **Manage Robot Controllers** field to add, remove, disable or modify robot controllers' data collection process.

- See [Chapter 6, Robot Configuration Commands](#) for information about creating commands.
- Click **Submit** to perform the operation.

Output is displayed in the bottom left field.

- Successful operations output is displayed in green.
- Failed operations output is displayed in red.

Note

Click **Export** to download a CSV file containing a history of configuration commands that were previously entered. See [Section 3.3.3, Export and Restore Data Collector Pull Schedule Records](#) for more information.

3.3.2 Most Recent Robot Operations

A history of recent operations is displayed in this field. Click an entry to display additional information.

Table 3.3.2 Most Recent Robot Operations Data Descriptions

Data	Description
Started By	The user who performed the operation.
Controller IP	The IP address of the robot that performed the configuration command. This information is available using the Robot Controllers window.
Operation	The operation that was performed.
State	That status of the operation, such as Pending or Done.

Data	Description
Result	The result of the operation, such as Success, Not Started, or Failed.

3.3.3 Export and Restore Data Collector Pull Schedule Records

3.3.3.1 Exporting Data Collector Pull Schedules

Click **Export** to save the current robot controller pull schedule to a comma separated value (CSV) file. This file contains a series of records that instruct the data collector when to retrieve a robot controller's data, and can be used to restore data collector pulling operations.

Supported Record Types

Two CSV record types are currently supported:

- Record type two (2) is for configuring and pulling legacy robot controllers.
- Record type seven (7) is for pulling ZDT robot controllers.

Exported Record Type Samples

Record Type: 2 - Legacy robot controller pull record

```
2,172.22.194.39,1,1,3,1.0,0,0,0,1,FRA,PDE,d1067b0d-8222-4049-8f59-1ca
be45f5b86,0
```

Record Type: 7 - ZDT robot controller pull record

```
7,172.23.194.164,3,1,0,0,0,BATT_LOW_T;GBRAKE_CHK_T;HOURLMETER_T;PCOD_D
AT_T;PROTECT_T;SVG_TUNE_T
```

3.3.3.2 Restoring Data Collector Pull Schedules

Restoring the exported file is different for legacy and ZDT robot controllers.

Restoring Legacy Robot Controller Pull Records

A file containing only legacy robot controller pull records can simply be loaded into the robot operations page. See [Chapter 6, Robot Configuration Commands](#), for more information.

Restoring ZDT Robot Controller Pull Records

A file containing ZDT robot controller pull records requires the following before the pull records will be accepted by the data collector.

3.3.3.2.1 Restoring ZDT Robot Controller Pull Records

About this task

Use this procedure to restore ZDT robot controller Pull Records.

Before you begin

You should have a CSV file from previous robot operations export procedure.

Procedure

1. Confirm that the robot controller is communicating with the data collector (using the data collector's robot controller's page). Each ZDT robot controller specified in the file (record type 7) must first be communicating with the data collector, or the restore will fail.
2. Remove any comment characters (#) that may be present at the start of the exported CSV record.
3. Use the CVS file to restore the pull records.
See [Chapter 6, Robot Configuration Commands](#), for more information.

3.3.3.3 Pull Schedules

Pull schedules display the frequency and time when data was pulled from legacy robots.

See [Chapter 6, Robot Configuration Commands](#), for information on defining a pull schedule.

Table 3.3.3.3 Pull Schedule Data Description

Data	Description
Robot Name	The device name
Controller IP	The device IP address
Controller ID	The device unique ID
Frequency Type	Defines if the data is pulled by minutes or other values
Frequency Value	How often the data is pulled
Day, Hour, Minute	Times at which the pull is scheduled
Last Pulled	When data was last pulled

3.3.4 Robot Controller File Backup

The ZDT Data Collector can be configured to automatically backup robot controller files on a daily basis and send the file archive to the cloud where it is then available for download through the ZDT Web Portal as needed. The robot controller file backup feature allows you to define a daily backup schedule, securely store backups in the cloud where they are available for download when required, identify and store a golden backup to protect the initial configuration, automatically store daily backups to protect the controllers' current configuration, and receive notifications if backups ever fail.

3.3.4.1 Robot Controller File Backup Configuration

About this task

Follow the following procedure to configure the data collector for robot controller file backup.

Before you begin

Complete the steps to provision your data collector and verify the connection in the **ZDT Web Portal**. Also verify that your network configuration will allow the data collector to connect to your robot controllers through FTP on port 22.

Procedure

1. Login to the **Data Collector Web UI** in your web browser
2. Go to the **Admin** screen by clicking the link on the side-navigation bar. Wait a couple moments for the status to change to enabled.

Controller Backup

Status: disabled

Enable Controller Backup

Figure 3.3.4.1 (a) Enable Controller File Backup Button

3. Login to the **ZDT Web Portal**
4. Click the Robot Backups link on the **Dashboard** to configure a backup schedule for your robots

Details

Search

 [Find Robot](#)

 [Robot Backups](#)

Figure 3.3.4.1 (b) Robot Backups Link - ZDT Web Portal

5. Once the backup schedule is configured, click the download button, at the bottom of the screen next to your data collector, to save the configuration as a csv file on your computer.
6. Go back to the **Data Collector Web UI** and click **ZDT Data Collector** on the side-navigation bar
7. Click on the **Robot Operations** tab at the top of the screen
8. Open the backup schedule csv file in notepad, you will see a line beginning with 107 for each robot
9. Verify that the robot ip addresses are correct. The data collector will use the ip addresses in the csv file to connect through FTP on port 22.
10. Copy and paste the contents of the backup schedule csv file from notepad to the input text box on the **Robot Operations** screen
11. Click the Submit button
12. Your data collector is now configured to backup your robots' controller files and send the backups to the cloud

What to do next

From the **ZDT Web Portal - Robot Backups** screen you will start to see your robot backups. From there, you can request a backup download link.

3.4 Message Stats

The Message Stats page displays how many messages were received from a robot, when a message was last received, how long it took for the message to be forwarded, and other information.

#	Message ID	Controller IP	Name	Line	Cluster	▼ Num Received	Last Received	Num Sent	Avg Send Time	Last Sent
1	VAROHL_T	192.168.1.100	Robot1	Line1	RoboGuide	30	2016-02-12T17:01:52	30	0.0574333333333333...	2016-02-12T17:01:52
2	VAROHL_T	192.168.1.100	Robot1	Line1	RoboGuide	20	2016-02-12T17:01:50	20	0.0566500000000000...	2016-02-12T17:01:51
3	VAROHL_T	192.168.1.100	Robot1	Line1	RoboGuide	20	2016-02-12T17:01:56	20	0.0530500000000000...	2016-02-12T17:01:56
4	VAROHL_T	192.168.1.100	Robot1	Line1	RoboGuide	19	2016-02-12T17:01:29	10	0.0514999999999999...	2016-02-12T17:01:29
5	VAROHL_T	192.168.1.100	Robot1	Line1	RoboGuide	11	2016-02-12T17:01:34	11	0.0520000000000000...	2016-02-12T17:01:34
6	VAROHL_T	192.168.1.100	Robot1	Line1	RoboGuide	10	2016-02-12T17:01:22	10	0.1494000000000000...	2016-02-12T17:01:23
7	VAROHL_T	192.168.1.100	Robot1	Line1	RoboGuide	10	2016-02-12T17:01:58	10	0.0528999999999999...	2016-02-12T17:01:58

Figure 3.4 Message Stats

This information is primarily for display, but you can also perform the following functions:

- **Reset Statistics** - Use this to get a better understanding of the information coming in live. For example, click Reset to retain all the rows but clear the number received, last received, number sent, average sent time and last sent time. The data will be updated, showing you what is occurring from that point.
- **Delete All Statistics** - Remove all message statistic rows. Only new data will be shown

Note

Enter a phrase or partial phrase in the Filter to display only robots that include that information. The filter is matched for all fields. For example, enter VAR to display only message names that include VAR. Enter a partial IP address to include devices with that root address.

Table 3.4 Message Stats Data Description

Data	Description
Message ID	The message name
Controller IP	The robot IP address
Name	The robot name
Line	A site specific name that can be assigned to the robot to help identify it.
Cluster	Cluster is a site specific name that can be assigned to the robot to help identify it.
Num Received	The number of messages received since the last reset
Last Received	When the last message was received
Num Sent	The number of messages sent on

Data	Description
Avg Send Time	The average time for the message to be processed and sent to the primary
Last Sent	When the last message of this type from this robot was sent

3.5 Log Files

Use **Log Files** to display logs for the processes that run on the server, for errors in robot operations, and other logs.

- Click a log file to display the content on that file.
- Log entries include when the log was created and the file size (in bytes).
- Use the browser's search tool to search for text within the log output.

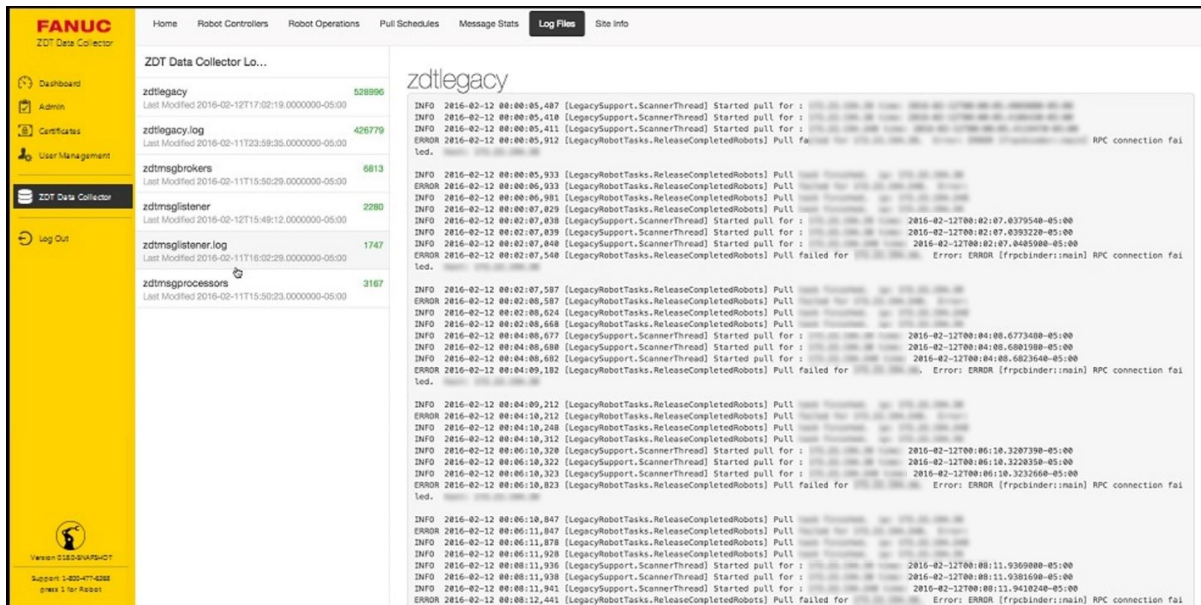


Figure 3.5 Log Files

Note

To refresh the log file display, click on another entry and then re-select the file. This prevents the log file from being dynamically updated while you are reviewing it.

3.6 ZXL Site Info (ZDT Provision)

This page displays data about the ZDT Data Collector, and the applications and libraries running on the system. Information from this page might be requested by FANUC support personnel when investigating an issue with the Data Collector.

FANUC ZDT Data Collector

Home Robot Controllers Robot Operations Pull Schedules Message Stats Log Files **Site Info**

Site Information Upload ZDT Provisioning File

Site ID	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
Data Collector ID	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
Data Collector IP	192.168.1.10
Legacy Listener Port	6062
Listener Port	6063
Customer ID	xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx
Country	USA
Last Updated	2016-04-26T16:47:15.000000+00:00

Application File Versions

File	Version	Modified Date
CommandLine.dll	1.0.0.0	2016-04-26T16:47:15.000000+00:00
log4net.dll	1.2.15	2016-04-26T16:47:15.000000+00:00
MySql.Data.dll	6.9.0	2016-04-26T16:47:15.000000+00:00
Newtonsoft.Json.dll	9.0.1	2016-04-26T16:47:15.000000+00:00
RabbitMQ.Client.dll	3.6.0	2016-04-26T16:47:15.000000+00:00
zdtDataAdapter.dll	1.0.0.0	2016-04-26T16:47:15.000000+00:00
zdtDataAdapter.XmlSerializers.dll	1.0.0.0	2016-04-26T16:47:15.000000+00:00
zdtDataArchiver.dll	1.0.0.0	2016-04-26T16:47:15.000000+00:00
zdtDataArchiver.XmlSerializers.dll	1.0.0.0	2016-04-26T16:47:15.000000+00:00

Version 1.4.0
Support: 1-800-477-6288
press 1 for Robot

Figure 3.6 Site Info

Table 3.6 (a) Site Information

Data	Description
Site ID	The globally unique ID of the site the ZDT Data Collector is monitoring.
Data Collector ID	The globally unique ID of this ZDT Data Collector
Data Collector IP	The server IP address
Legacy Listener Port	6062 (unless it has been reconfigured for special circumstances)
Listener Port	6063 (unless it has been reconfigured for special circumstances)
Customer ID	The customer's globally unique ID
Country	The country where the collector is installed
Last Updated	The date/time that the site was last updated

Additional data is displayed for the library files used for the Application, Web, and Web API files.

Table 3.6 (b) Application, Web, and Web API File Versions

Data	Description
File	The application file name
Version	The version number of the application file
Modified date	The date/time stamp of the application

4 User Management

User Management allows administrators to provide access to additional LDAP users. You can also disable or re-enable the default admin account.

Related concepts

[Change Disable or Enable the Admin Password](#)

[Configure the LDAP Server Connection](#)

[Manage User Access](#)

[User Roles](#)

4.1 Change Disable or Enable the Admin Password

A default admin account allows you to perform the initial setup and add users to the ZDT Data Collector. This is the only local account provided by the ZDT Data Collector.

4.1.1 Logging In Using the Default Admin Password

Default admin password

About this task

Use this procedure to log in using the default admin password.

Before you begin

URL is known.

Procedure

1. Open a web browser and enter the URL provided by your support representative.
2. Log in using the default administrator password.

The default credentials are:

Username: admin

Password: admin

4.1.2 Changing the Admin Password

Change the admin password

About this task

Change the default admin password after the first login to prevent unauthorized users from accessing system information and data

Before you begin

Current admin user password is known.

Procedure

1. Log in as the admin user.
2. Click **Admin**.
3. Enter the current admin password.
4. Enter and re-enter a new password.
5. Click **Change Password**.

4.2 Configure the LDAP Server Connection

Users added to the ZDT Data Collector must have accounts on the LDAP servers. The LDAP server provides user authentication, including passwords for each user.

You can add multiple LDAP server groups, if necessary.

The screenshot shows the FANUC ZDT Data Collector web interface. On the left is a yellow sidebar with navigation links: Dashboard, Admin, Certificates, User Management (highlighted), Network Configuration, ZDT Data Collector, and Log Out. At the bottom of the sidebar is the ZDT logo and version information: Version 0.21.0-SNAPSHOT and Support: 1-800-477-6268 press 1 for Robot. The main content area has a top bar with 'User Roles' and 'LDAP Configuration' tabs. Below this is the title 'New LDAP Server Group'. The configuration fields include: 'Enabled' (checkbox), 'SSL Required' (checkbox), 'Authentication Method' (radio buttons for 'Direct' and 'Search', with 'Direct' selected), 'Alias' (text input with 'example-ldap-server-group'), 'User Base DN' (text input with 'dc=example,dc=com'), 'User RDN Attribute' (text input with 'e.g. uid'), 'Hostname' (text input), and 'Port' (text input with a dropdown arrow). At the bottom right is a yellow '+' button. At the bottom left are 'Save' and 'Cancel' buttons.

Figure 4.2 LDAP Configuration

4.2.1 Configuring LDAP

About this task

Use this procedure to configure LDAP.

Procedure

1. Click **User Management > LDAP Configuration**.
2. Click **Add LDAP Server Group**.
3. Select **Enable** to allow users on an LDAP server to access the ZDT Data Collector.
4. (Optional) Select **SSL Required** to enable encrypted (secure) communication with the LDAP server. This allows the Data Collector to securely authenticate user logins with the LDAP server provided for the site.
 - a) Select **SSL Required**.
 - b) Upload an LDAP Server Certificate (under **Certificates > LDAP**) to install a trusted certificate for the LDAP server. See [Section 5.1, Upload the Required Security Certificates](#) for instructions.
5. Select an authentication method and enter the required server settings.
 - *Direct* - attempts to authenticate using a DN built from the submitted username in the form of `userRdnAttribute}={username},{userBaseDn}`. See [Table 4.2.1 \(a\)](#) for setting descriptions.

Table 4.2.1 (a) Direct Settings

Setting	Description
Alias	A meaningful name for the server cluster Example: LDAP server group 1
User Base DN	A set of fields and values that are common to all users' Distinguished Names that should be able to login to ZDT. For example: <code>dc=example,dc=com</code>
User RDN Attribute	User ID Attribute—The name of the LDAP mapping field where the User ID is stored. For example: <code>uid</code>
Hostname	The server hostname or IP address.
Port	The server port. – Port 389 is typically used for LDAP communication. – Port 636 is typically used for connecting to an LDAP server over SSL.

- **Search** - attempts to authenticate using a “search user” account, then searches for a user object that contains the submitted username. If one is found, it then attempts to authenticate using that user object's DN in the form of `((objectClass={userObjectClass}))({userIdAttribute}={username}))`. See [Table 4.2.1 \(b\)](#) for setting descriptions.

Table 4.2.1 (b) Search Settings

Setting	Description
Alias	A meaningful name for the server cluster Example: LDAP server group 1
Search User DN	The fully qualified Distinguished Name of a user that is able to query the LDAP server. This will be used to find the full DN of the user logging in. Example: cn=read-only-admin,dc=example,dc=com
Search User Password	The search user password.
Search Base DN User Object Class User ID Attribute	These fields are used to create the search query to find the user (together with the username entered in the login screen). – User Object Class - The name of the LDAP server attribute that holds the user's name. – User ID Attribute - The name of the LDAP mapping field where the User ID is stored. For example: uid – Search Base - A set of fields and values that are common to all user's Distinguished Names that should be able to login to ZDT. For example: dc=example, dc=com
Hostname	The server hostname or IP address.
Port	The server port. – Port 389 is typically used for LDAP communication. – Port 636 is typically used for connecting to an LDAP server over SSL.

6. (Optional) Click + to add additional LDAP servers to the server group. Additional LDAP servers will only be used if connecting to the first LDAP server fails.
7. Click **Submit** to save the changes.
8. (Optional) Repeat these steps to add additional LDAP server groups.

4.3 Manage User Access

After configuring LDAP, you must add a user and assign them one or more roles before they will be able to login to the ZDT Application. The username given here must exactly match the username in LDAP.

Add additional LDAP users, if necessary. Each user is assigned one or more roles, which defines their access to the available features and windows.

- See [Section 4.3.1, Adding Users](#).
- See [Section 4.3.2, Modifying User Access Roles](#).

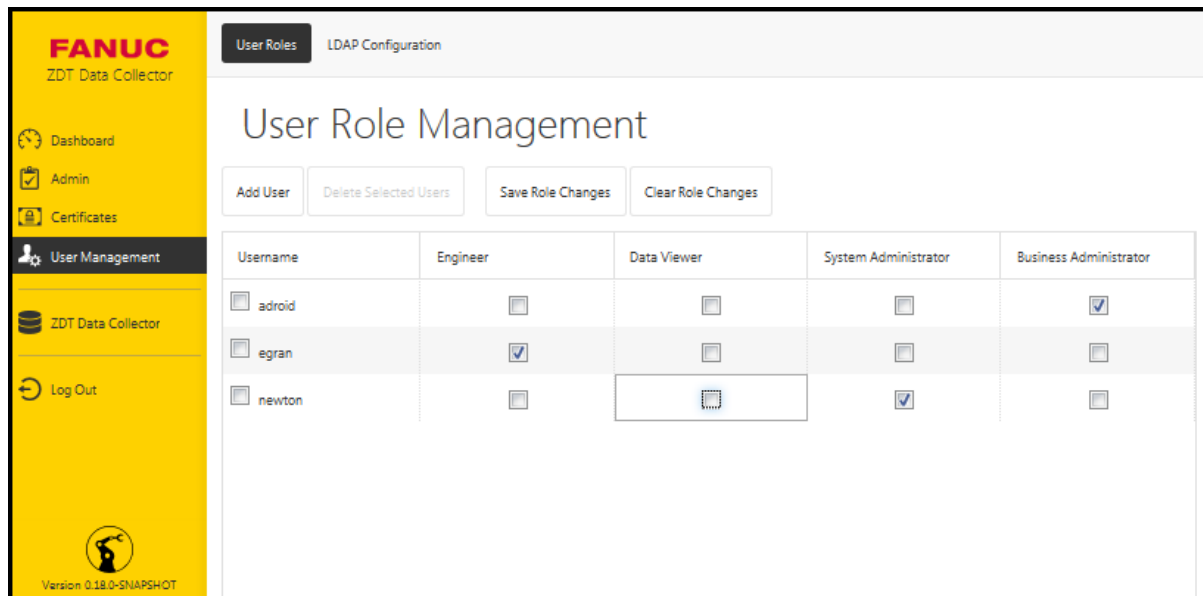


Figure 4.3 User Role Management

4.3.1 Adding Users

The User Roles assigned to a person define their ability to view and access the various features and pages in the ZDT Data Collector.

About this task

Use this procedure to add users.

Before you begin

Users must be previously configured on an LDAP server.

Procedure

1. Click the **User Management** tab.
2. Click **Add User**.
3. Enter an existing LDAP username.
4. Select one or more user roles. [Section 4.4, User Roles](#) describes the access privileges for each user role.
5. Click **Add User**.
6. Add additional users as necessary.
7. When done, click **Cancel** to close the window.

4.3.2 Modifying User Access Roles

Use this procedure to change user access roles.

About this task

Use this procedure to modify user access roles.

Before you begin

Users must be previously configured on an LDAP server.

Procedure

1. Click the **Add Management** tab.
2. Modify the user roles for one or more users.
[Section 4.4, User Roles](#) describes the access privileges for each user role.
3. Click **Save Role Changes**.
4. Click **Save**.
[Section 4.4, User Roles](#) describes the access privileges for each user role.

4.4 User Roles

The following user roles define the features and windows that can be accessed by each user. Click **User Management** to add or revise users.

Table 4.4 User Roles

Page	Action	Admin (non-LDAP user) ¹	Data Viewer	Engineer	System Administrator ²	Business Administrator ³
Dashboard	View All Information	Allowed	Allowed	Allowed	Allowed	Allowed
Admin	View All Information	Allowed	Allowed	Allowed	Allowed	Allowed
Admin	Update Satellite	Allow	Denied	Denied	Allowed	Allowed
Admin	Download Logs	Allowed	Denied	Allowed	Allowed	Allowed
Admin	View Disable/Enable Admin User Button	Denied	Denied	Denied	Allowed	Denied
Admin	View change admin password	Allowed	Denied	Denied	Denied	Denied

¹ The non-LDAP admin is the “admin” account used to sign in before LDAP is set up. This account can be disabled as described in [Section 4.1, Change Disable or Enable the Admin Password](#).

² The system administrator is an LDAP user role that provides full privileges.

³ The business administrator is an LDAP user role that includes most admin privileges, such as the ability to add/modify user roles. However, the business admin cannot modify the system administrator role.

Page	Action	Admin (non-LDAP user) ¹	Data Viewer	Engineer	System Administrator ²	Business Administrator ³
Admin	Update Time Zone	Allowed	Denied	Denied	Allowed	Denied
Admin	Restore	Allowed	Denied	Denied	Allowed	Denied
Admin	Backup	Allowed	Denied	Denied	Allowed	Denied
Certificate Management	Update All Certs	Allowed	Denied	Denied	Allowed	Denied
User Management	View and Change All Information	Allowed	Denied	Denied	Allowed	Allowed
ZDT Data Collector	View All Information	Allowed	Allowed	Allowed	Allowed	Allowed
ZDT Data Collector	Perform Robot Operations	Allowed	Denied	Allowed	Allowed	Allowed
ZDT Data Collector	Clear Message Stats	Allowed	Denied	Allowed	Allowed	Allowed
ZDT Data Collector	Reset Message Stats	Allowed	Denied	Allowed	Allowed	Allowed
ZDT Data Collector	Restart Processes	Allowed	Denied	Denied	Allowed	Denied

¹ The non-LDAP admin is the “admin” account used to sign in before LDAP is set up. This account can be disabled as described in [Section 4.1, Change Disable or Enable the Admin Password](#).

² The system administrator is an LDAP user role that provides full privileges.

³ The business administrator is an LDAP user role that includes most admin privileges, such as the ability to add/modify user roles. However, the business admin cannot modify the system administrator role.

5 Administration Tasks

Use the **Admin** and **Certificates** pages to manage updates to the system software, manage security certificates, and download system logs.

Related concepts

[Upload the Required Security Certificates](#)

[System Software](#)

[Backup and Restore the Satellite Configuration](#)

Related tasks

[Downloading System Logs](#)

[Changing the Satellite Time Zone](#)

5.1 Upload the Required Security Certificates

The following security certificates are required to provide secure communication between the ZDT Data Collector, robots, cloud services, and other services.

Note

All of the following security certificates are required for proper operation and data reporting on the ZDT Data Collector. Most of these certificates should be properly installed on installation, but you will have to install the HTTP Server Certificate as part of the initial setup. You can then selectively install new versions, if needed (see [Section 5.1.1, Installing the Certificates](#)).

Table 5.1 Required Security Certificates

Security Certificate	Description
HTTPS Server Certificate	(Recommended) Allows for secure communication with the web console UI and the Satellite REST API. This certificate is unique for each Data Collector and must be generated by the IT department for the site in which it is operating. This certificate must have the domain name of the ZDT Appliance as one of the Common Names (CN) in the subject field. It also must be signed using at least a SHA256 hashing algorithm. If not, users may get security warnings when visiting the ZDT Application web UI.

Security Certificate	Description
Trusted SSL Client Certificates	(Optional) Trusted SSL Client Certificates are certificates used by the server to communicate with trusted clients. Example clients are the on-box ZDT client and customer specific monitoring clients. These certificates allow the Satellite to trust clients to make REST API requests using SSL authentication. You can upload multiple Trusted SSL Client Certificates: One for each client that will need to use SSL authentication for API requests.
Satellite-to-Primary Client Certificate	(Required) Allows the Data Collector to securely send the messages to the Cisco Cloud. This file provides a private key and certificate chain to identify the ZDT Appliance to the Cisco Cloud, allowing it to send data to the cloud. FANUC/Cisco will provide this if a replacement is required.
Trusted Satellite-To-Primary Server Certificate	(Required) The Trusted Satellite-To-Primary Server Certificate is the public key of the Primary server in the cloud. The satellite uses this certificate to ensure it is talking to a trusted Primary server.
LDAP Server Certificate	(Optional) Enables encrypted (secure) communication with the LDAP server.

Note

An additional LDAP trusted certificate is required to allow the Data Collector to securely authenticate user logins with the LDAP server provided for the site. See [Section 4.2, Configure the LDAP Server Connection](#) for more information.

5.1.1 Installing the Certificates

About this task

Use this procedure to Install the Certificates

Procedure

1. Obtain the security certificate file(s) from your support representative.
2. Click the **Certificates** tab.
3. Replace the HTTPS Server Certificate. See [Figure 5.1.1 \(a\)](#).
 - a) Click the **API** tab.
 - b) Click **Replace HTTPS Server Certificate**.
 - c) Click **Browse** and select the file from a local drive.
 - d) Click **Upload** and then **OK**.

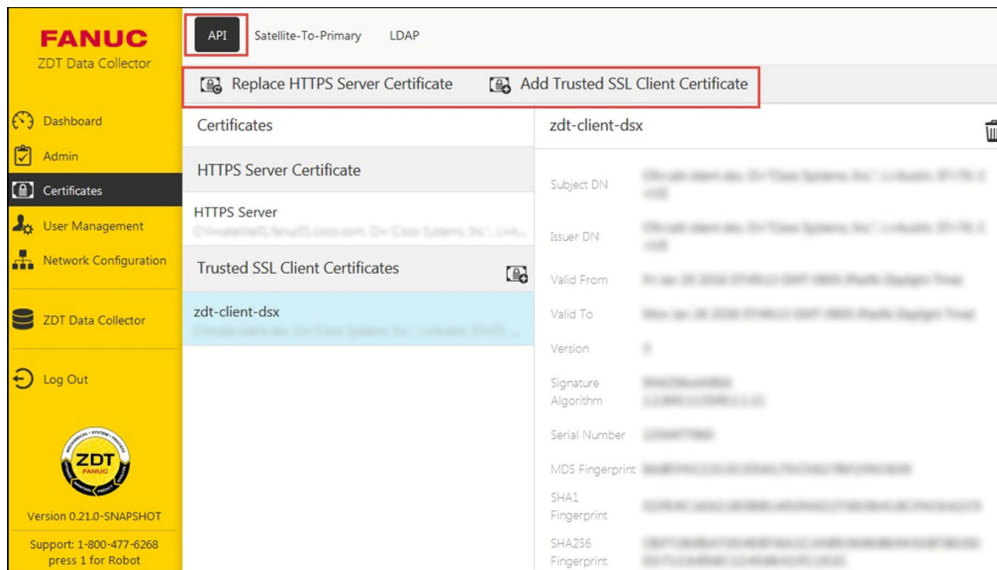


Figure 5.1.1 (a) API Certificates Tab

4. **Add the Trusted SSL Client Certificate.** See Figure 5.1.1 (a).
 - a) In the API tab, click **Add Trusted SSL Client Certificate**.
 - b) Click **Browse** and select the file from a local drive.
 - c) Click **Upload** and then **OK**.
5. **Replace the Satellite-To-Primary Client Certificate.** See Figure 5.1.1 (b).
 - a) Click the **Satellite-To-Primary** tab.
 - b) Click **Replace Satellite-To-Primary Client Certificate**.
 - c) Click **Browse** and select the file from a local drive.
 - d) Click **Upload** and then **OK**.

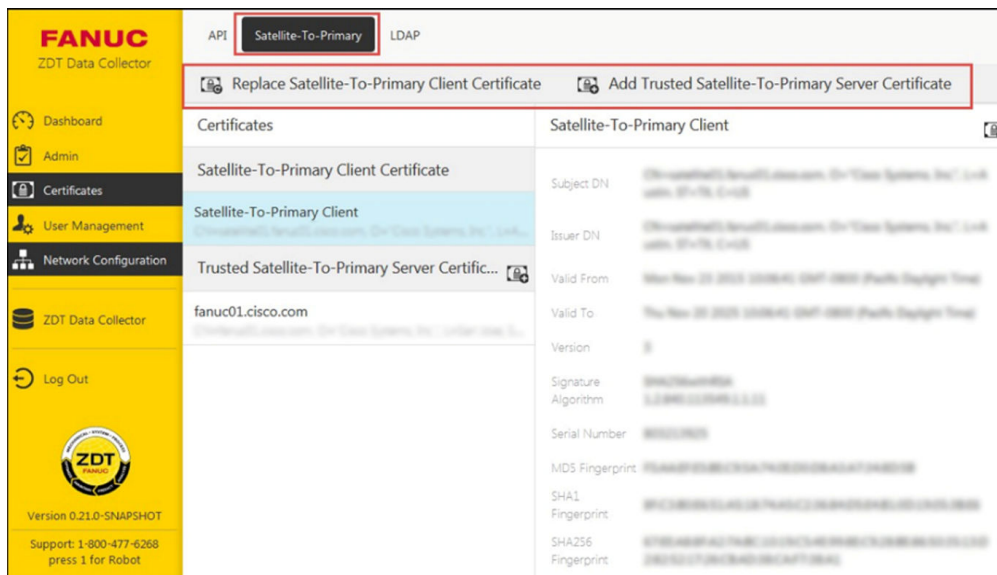


Figure 5.1.1 (b) Sattelite to Primary Certificate Tab

6. **Replace the Trusted Satellite-To-Primary Server Certificate.**
 - a) Click **Add Trusted Satellite-To_Primary Server Certificate**.
 - b) Click **Browse** and select the file for Trusted Satellite-To-Primary Server Certificate.
 - c) Enter an Alias to help you remember this certificate.

- d) Click **Upload** and then **OK**.
7. Upload the LDAP Server Certificates, if required. See [Figure 5.1.1 \(c\)](#).

LDAP certificates are required if the SSL option is selected in the LDAP server configuration (see [Section 4.2, Configure the LDAP Server Connection](#)).

- a) Click the **LDAP** tab.
- b) Click **Add Trusted LDAP Server Certificate**.
- c) Click **Browse** and select the file from a local drive.
- d) Click **Upload** and then **OK**.

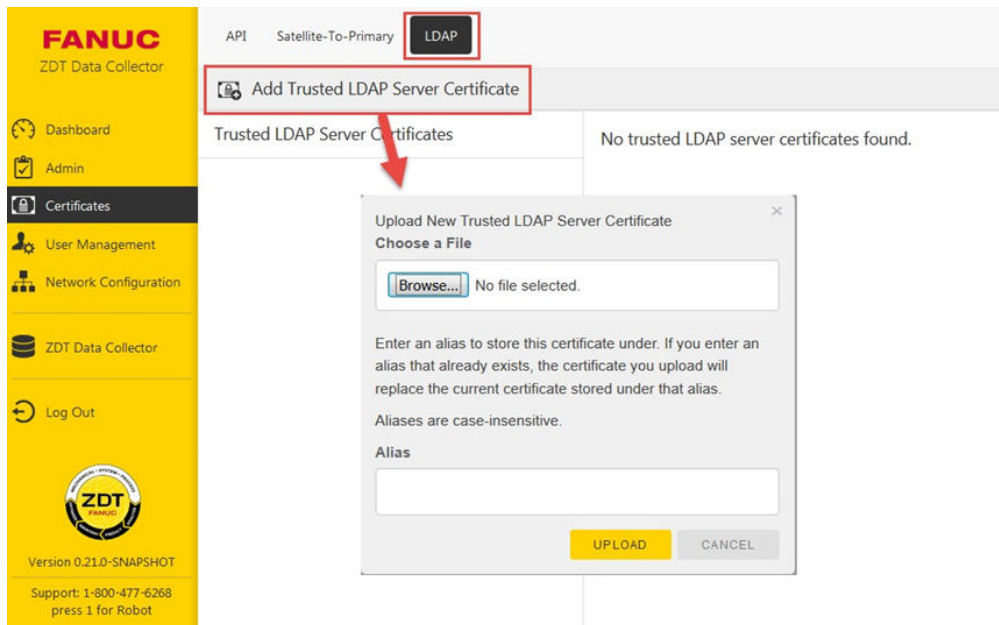


Figure 5.1.1 (c) LDAP Certificate Tab

5.2 Downloading System Logs

To download system logs for troubleshooting system issues, perform the following procedure.

About this task

Use this procedure to download system logs for troubleshooting system issues.

Procedure

1. Click **Admin** and then **Download Logs Archive**.
2. Click **OK** when the download is complete.
3. Access the file in your local download folder.

Note

If the message `This file couldn't be downloaded.` appears when using IE 11, please install the latest Microsoft security patch. See [Section 1.1, Requirements](#) for more information.

5.3 System Software

Use the Satellite options to view information about the ZDT Data Collector system, and to update the system software. For example:

- *Satellite Version*—The running version of ZDT Data Collector system software.
- *Last Updated*—The date and time when the system software was last updated.
- *Uptime*—The amount of time that the system has been running (in days:hours:minutes:seconds).

5.3.1 Updating the System Software

About this task

Use this procedure to update the system software.

The ZDT Data Collector UI or functions can be updated using a signed image provided by your support representative.

Before you begin

The update file must be a valid format that includes the correct security key.

Procedure

1. Click **Admin** and then **Update**.
2. Select a file from a local or network drive.
3. Click **Upload**.
4. Click **OK** when the upload is complete.

5.4 Changing the Satellite Time Zone

About this task

Use this procedure to change the Satellite Time Zone.

Procedure

1. Under **Satellite**, click the pen symbol next to **Time Zone**.
2. Select a new setting and click **Update**.

5.5 External Queue (Add-On feature)

The **External Queue** option will show up if you purchased the external queue add-on feature. This add-on option, if enabled, allows access to the robot messages from Data Collector RabbitMQ external message queue. RabbitMQ API is required to read the messages on this external queue. Please contact your support representative on how to purchase this add-on feature and external queue configuration.

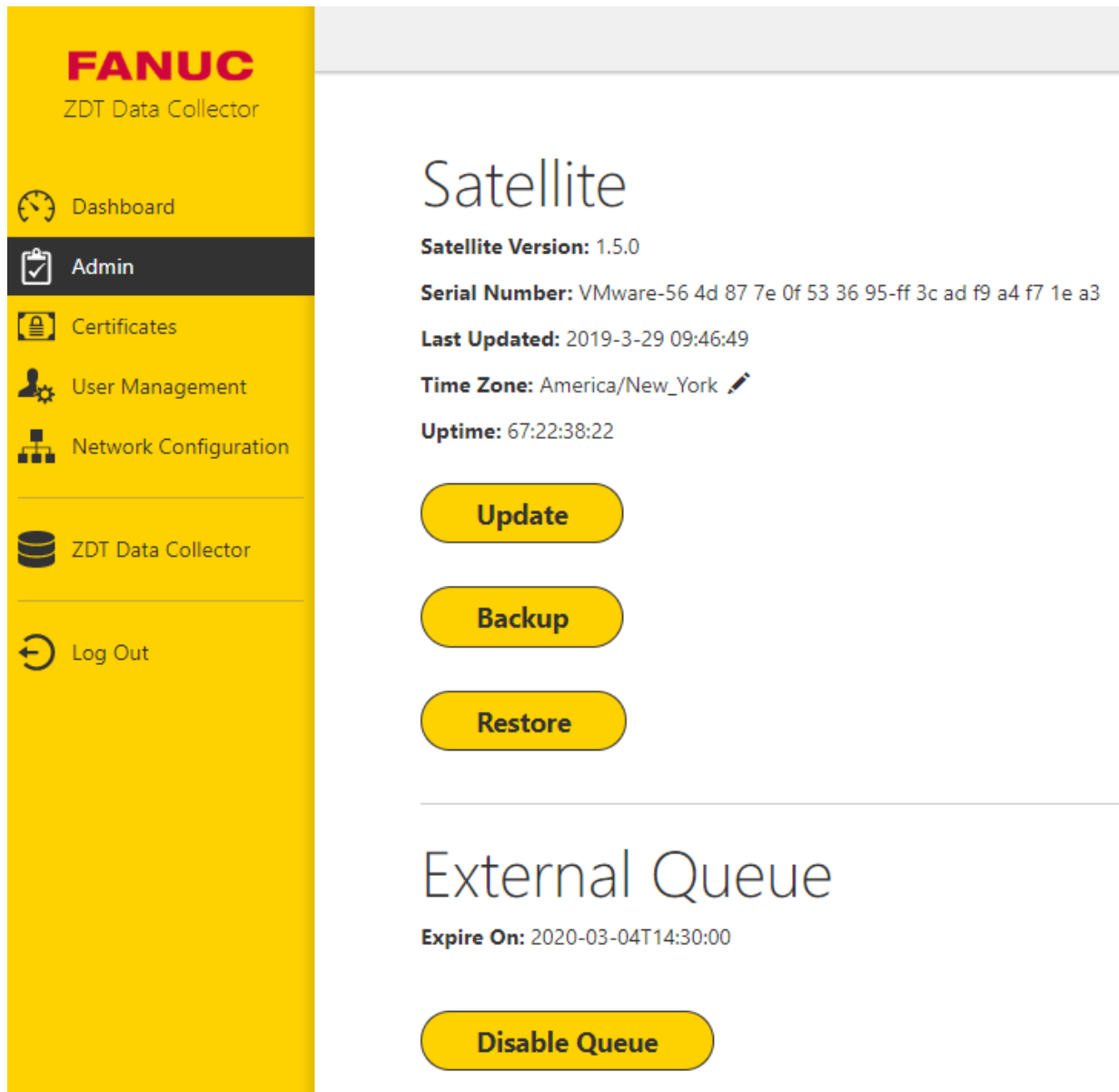


Figure 5.5 External Queue Enable/Disable

5.6 Decommissioning a Data Collector (Optional)

If for some reason a Data Collector must be decommissioned, three operations must be performed in the following order.

Note

If you are decommissioning because you are **replacing** OR **upgrading** your Data Collector, skip directly to [Section 5.6.4, Shut Down the Data Collector](#).

1. Obtain the CSV file list of all robots currently communicating to the data collector. This procedure should be done by the location's I.T. department.
2. Disable the ZDT Client from the robot controller teach pendant. Repeat this procedure for each robot controller in the CSV file list obtained in operation 1 above.

3. Shut down the data collector from VMWare ESXI. This procedure should be done by the location's I.T. department.

5.6.1 Obtain the CSV List of all Robots Communicating to the Data Collector

About this task

This procedure should be performed by the I.T. department.

Procedure

1. Log in to the data collector's Web UI.
2. From the left-hand menu, select **ZDT Data Collector**.
3. From the top tab bar, select **Robot Controllers**.
4. Press the **CSV Export** button in the top left of the Robot Controller list.

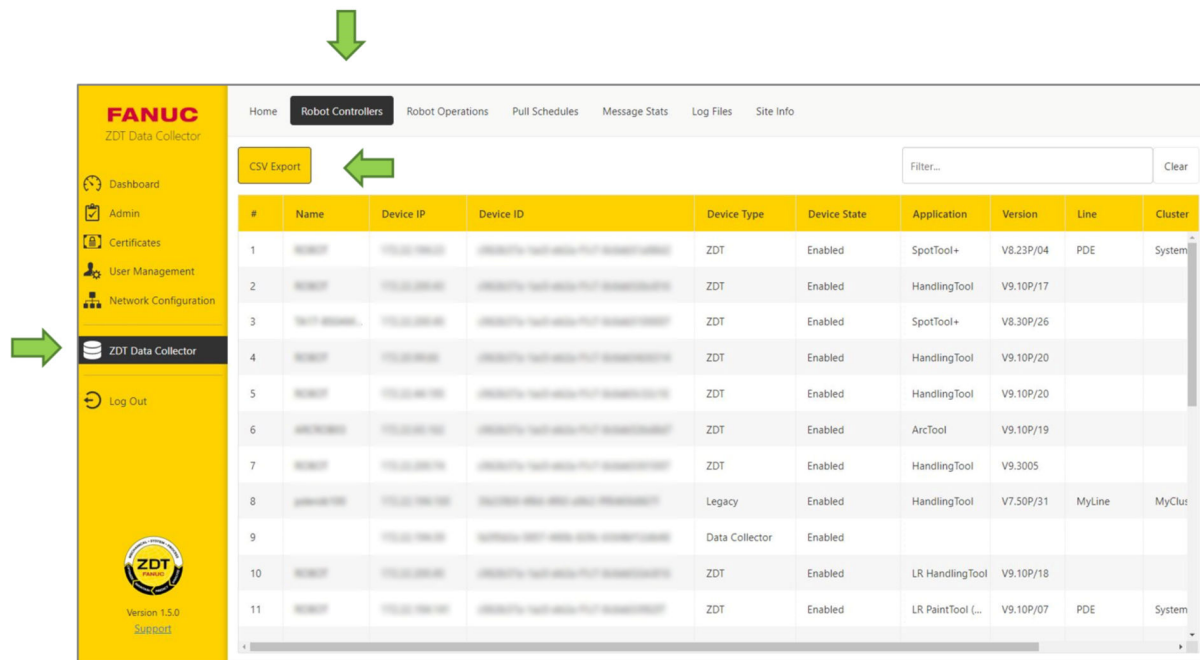


Figure 5.6.1 CSV List

5.6.2 Disable ZDT on a ZDT Robot Controller from the Teach Pendant

About this task

Use this procedure to disable ZDT from the Teach Pendant.

Procedure

1. Robots listed as device type "ZDT" need to have ZDT disabled at the robot. On the robot Teach Pendant press **MENU > SETUP > ZDT Client** to go to the ZDT Client Setup screen.

Note

Older robot software versions may not have the ZDT Client setup menu selection. In this case, follow [Section 5.6.3, Delete Legacy Robot Controller on the Data Collector](#) to disable ZDT.

2. Select **ZDT Client**.
3. Press **F5 [DISABLED]**.

5.6.3 Delete Legacy Robot Controller on the Data Collector

About this task

Use this procedure to delete a legacy robot controller on the Data Collector.

Procedure

1. Robots listed as device type `Legacy` need to have ZDT disabled remotely from the data collector.
2. On Data Collector Web UI, go to **ZDT Data Collector > Robot Operations** (See [Section 3.3, Robot Operations](#)).
3. Issue operation 4 command to disable Legacy robot controller on the Data Collector. E.g. 4, 192.168.123.100
4. Click **Submit**.

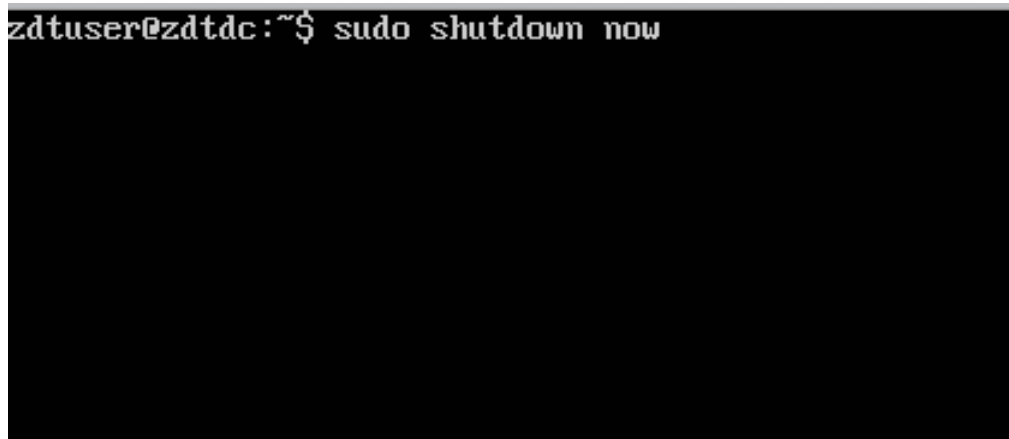
5.6.4 Shut Down the Data Collector

About this task

This procedure should be performed by the I.T department.

Procedure

1. Log in to the Data Collector VM through VMWare ESXI, not the Web UI.
2. After logging in to the VM, select **Virtual Machines** in the Navigator Window.
3. Select the data collector from the VM list.
4. Click the **Console** button.
5. Select **Open Console Browser**. A command window will open.
6. Log in using Data Collector VM password.
7. At the command prompt type `Sudo shutdown now` and press **[Enter]**.



```
zdtuser@zdtcd:~$ sudo shutdown now
```

Figure 5.6.4 Sudo shutdown now Message

8. Wait for the shutdown process to complete.
9. Power off Data Collector VM through VMWare ESXI.

6 Robot Configuration Commands

A robot configuration must be added to the ZDT Data Collector for it to handle messages from G1–V7.20 or G2 - V7.50 robots.

Robot controllers are added to the ZDT Data Collector using a comma-separated value (CSV) file. This CSV file format is similar to the DiagnosticPRO CSV file format, but includes additional fields not supported by DiagnosticPRO.

For example:

```
0,1,,,,,,,,,
1,IP,Job_Data,Condition_Data,Frequency_Type, Frequency
Number,D,H,M,MRA,Line,Cluster,Controller_ID,Disable
2,10.22.58.167,1,1,1,1,0,14,0,1,Line1,Cluster1,80d63113-60e5-44cc-
8bc6-a57bcb0b8ec6,0
```

Note

The first two lines are optional.

6.1 CSV Column Definitions

This table describes the columns included in the CSV file.

Table 6.1 CSV Column Definitions

Column ID	Description
(first column)	0 = File format version 1 = Header labels 2 = Add robot controller 3 = Delete the robot (only the IP is required) 4 = Disable the robot (only the IP is required) 5 = Enable the robot (only IP is required) 6 = Pull robot controller now (only IP is required) 7 = Add robot controller pull schedule
IP	IPv4 address of the robot controller being added
Job_Data	0 = Disable push data 1 = Enable push data
Condition_Data	0 = Disable pull data 1 = Enable pull data
Frequency_Type	0 = Hourly pull trigger, 1 = daily pull trigger, 2 = weekly pull trigger
Frequency Number	Number of hours/days/weeks (depending on type above) between issuing the pull.
Column ID	Description

Column ID	Description
D	Day of the week for weekly pull operation to trigger: (0=Sunday, 1=Monday, 2=Tuesday, 3=Wednesday, 4=Thursday, 5=Friday, 6=Saturday)
H	Hour for daily/weekly pull operation to trigger. The vales are 0-23 (0 = midnight)
M	Minutes after the hour for daily/weekly pull operation to trigger. The vales are 0-56
MRA	Enable collection of servo-off detail data. 0 = Disable servo-off detail file collection 1 = Enable servo-off detail file collection
Line	Name of the line that the robot is on. This is a first level user defined string to help organize robot controllers in a facility.
Cluster	Name of the cluster that the robot is in. This is a second level user defined string to help organize robot controllers in a facility.
Controller_ID	Controller GUID
Disable	(Optional) An additional field is used to disable the robot when adding it to the data collector. This field is typically used only when restoring from a backup if the robot controller was disabled when the backup occurred.

6.2 Adding a Robot Configuration

About this task

Use this procedure to add a robot configuration.

Before you begin

Make sure the Data Collector is operational and the target robot controller is connected to the network.

Procedure

1. Create a CSV fole using the descriptions in [Table 6.1](#).
2. Select **Robot Operations**.
3. Copy the text of the CSV file and paste it into the **Manage Robot Controllers** box.
4. Click **Submit**.

Note

G3+ - V8.XX robots do not need to be configured. The Data collector automatically adds G3+ robots when messages are received from the robot.

7 Troubleshooting

7.1 Initial Setup

These are issues you might encounter when initially installing the data collector.

7.1.1 The VM console displays **Visit this URL in your browser to continue your set-up: https:// without IP address**

Cause

The VM's network configuration is invalid.

Remedy

Procedure

1. If DHCP is being used, check to make sure the DHCP server is functioning properly and then reboot the VM.
2. If this VM requires a static network configuration, a reinstallation may be required so that the network can be setup properly on first boot.

7.1.2 **This site cannot be reached is displayed when the console is launched for the first time after the .OVA is deployed**

Cause

It is possible that the PC you are using is not on the same subnet as the data collector.

Remedy

Procedure

1. Try pinging the IP address from the PC.
2. Make sure your PC has access to the data collector.

7.1.3 Your Connection is not Private appears when launching the web browser to access Data Collector Web UI

Cause

The default HTTPS certificate is not customized for your organization.

Remedy

Procedure

1. Click the **Advanced** link.
2. When a new link with the working **Proceed to xxx.xxx.xxx.xxx (unsafe) displays**, click it.

7.1.4 There was an internal server error is displayed when logging on

Cause

The internal server error is displayed if LDAP is not configured properly.

Remedy

Procedure

1. Try logging in with your company login ID and password.
2. Double check all the settings.
3. Verify that the specified LDSP server can be reached on the specified port.

7.1.5 The top four processes stopped on the ZDT Data Collector Home page

Cause

The ZDT provisioning file has not been properly applied.

Remedy

Procedure

1. Go to the ZDT Data Collector **Site Info** page.
2. Upload the ZDT provisioning file supplied by FANUC.

7.2 Robot Operations

These are issues you might encounter when managing the connection between the robots and the data collector.

7.2.1 How do I know that a robot has been properly configured?

When a CSV line has been submitted on the Robot Operations page, some basic validation of the line will be performed. The results of this check will be immediately shown in the box below. If there are any red status lines, the whole CSV submission will have been rejected. You must fix these and resubmit.

When the CSV submission has been accepted, a new status for each robot will be displayed in the grid to the right. All will initially show a `not Started` status. After a two to five minute delay, all status lines should go to `Success` or `Failed`. Click on the line for which you want more detail. A typical failure is that the robot is not powered up. You do not have to resubmit the CSV line for this robot. The data collector will retry the robot each time it is scheduled to pull from it.

If you made a mistake in the IP address, the data collector will try forever and never connect to the robot. To correct this, submit a new CSV line that deletes the robot.

7.2.2 How can I discover robots that have not responded to a pull in a long time?

Navigate to the ZDT Data Collector page and select the **Pull Schedules** tab. Click on the column header entitled **Last Pulled**. The list of robots will be re-ordered by the times indicated in this column.

You can issue an immediate pull by submitting a Robot Operation using a CSV line with the number six in the first column and just the IP address in the second.

7.2.3 What does **ERROR [frpcbinder::main] RPC connection failed. host: xxx.xxx.xxx.xxx in the detailed operation status mean?**

When you add or modify a robot pull schedule, the data collector tries to adjust some settings on the robot itself. If the robot is powered down, the data collector will not be able to make the settings at the time you made the change. The data collector will make further attempts at the times you have identified to pull data. When the robot is finally powered up, the data collector will successfully configure the robot and begin pulling data.

7.2.4 Why are the Name, Application, and Version entries blank for a specific-robot?

These details are filled in the first time the data collector can successfully pull a legacy robot controller or when a ZDT robot controller successfully sends its controller information message. Blank entries mean that the data collector has not successfully pulled or received any messages yet. Verify that the robot is connected and operating properly.

7.2.5 Why are no GJOB_DATA_T messages coming from a specific R30iA controller?

There are several things to check:

- Check the robot controller's operation in the ZDT Data Collector UI to verify that there was no error in configuring the robot.
- Make sure the robot system variables are set using the teach pendant.
 - Server name matches the IP address of the data collector reported on the site info page.
 - Port number matches the value reported on the site info page.
 - Make sure collection is enabled.

`$CONDET_CFG.$ENABLE = 1`

- Make sure the group is enable.

`$CONDET_GRP[x].$ENABLE = 1`

- Make sure the group is setup to pass some type of high water mark data.

`$CONDET_GRP[x].$DATA_TYPE = 7 (111) for all data.`

- Check the listener log file and the robot operations log for issues.

7.2.6 Why are no Pull messages coming from a specific R30iA controller?

There are several things to check:

- Check if the robot has a pull schedule defined.
- Check when the robot's data was last pulled via the pull schedule page.
- Check the last robot operation performed on that robot and its results.
- Check the message statistics screen to see if the robot has communicated with the data collector and whether the data has been sent to the FANUC data center.
- Check the listener log file and the robot operations log for issues.

7.2.7 Why are no servo-off detail messages coming from a specific R30iA controller

There are several things to check:

- Check if the robot has a pull schedule defined.
- Check when the robot's data was last pulled via the pull schedule page.
- Check the message statistics screen to see if the robot has communicated with the data collector and whether the data has been sent to the FANUC data center.
- Check the legacy log file to verify there were no error with retrieving the servo-off detail data.

– The log file will report whether the memory card might be full or missing.

`$SERV_DEV` indicates the device where servo-off details files are saved.

– The log file may report that the robot controller has servo-off detail collection disabled.

`$SERV_OUTPUT` must have a value of 0x10 to enable collection of the servo-off detail records. Each servo-off detail has 10 records. Therefore, it takes 11 servo-off events to generate one file.

7.2.8 Why are no ZDT messages coming from a specific R30iB controller?

There are several things to check:

- Check the message statistics screen to see if the robot has communicated with the data collector and whether the data has been sent to the FANUC data center.
- The robot controllers must be power cycled when changes are made to the ZDT Client configuration (RCMCFG.XML).
- Verify the **Setup ZDT Client** screen data on the TP.
 - Verify the URL has the correct IP address of the data collector as shown on the site info page.

Data Collector IP (Make sure the robot controller has been assigned a static IP).

- Make sure the port number is correct as shown on the site info page.

Listener Port (default is 6063).

- The URL is case sensitive so make sure the case of the URL is correct. The data collector expects the following case of the URL ZDTData.

<http://xx.xx.xx.xx:6063/ZDTData>.

- Verify the **Host Comm** screen data on the TP.

Make sure controller IP address, subnet mask and router IP are set up correctly.

7.3 Dashboard Status

These are actions you might take when the dashboard indicates a problem.

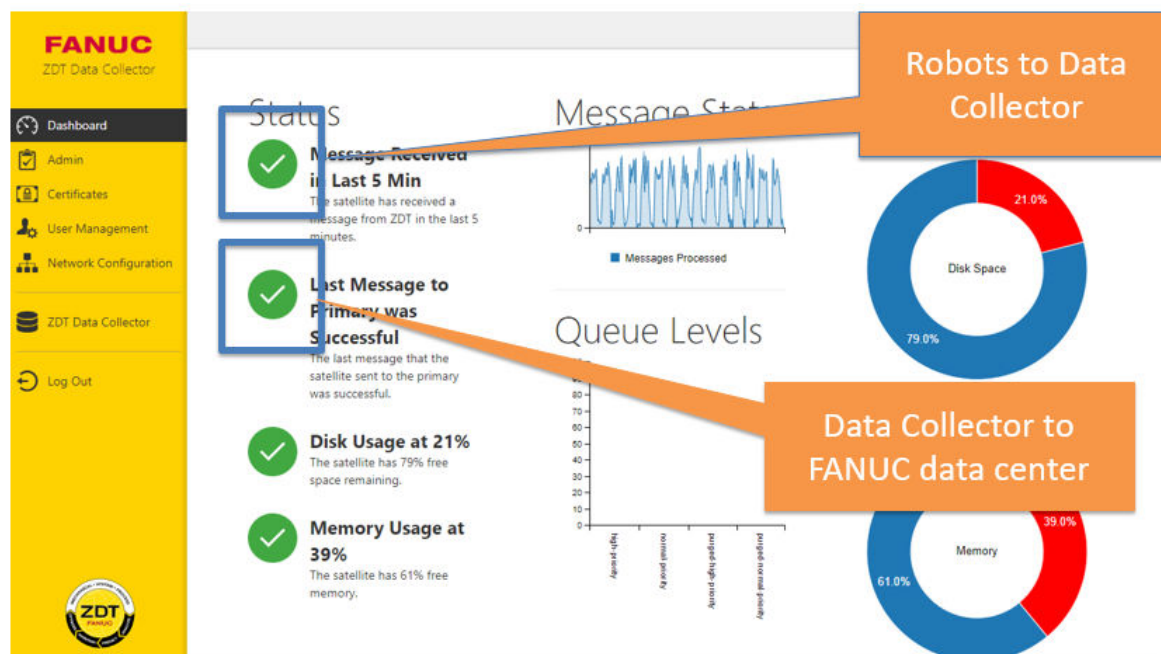


Figure 7.3 Dashboard Status

7.3.1 What if the Message Received in Last five Minutes is Red?

Green indicates Data Collector received messages from the robots in the last five minutes. It will remain red until you provision Data Collector with ZXL site info file (required). Robots do not send messages to Data Collector all the time. When testing with one or few robots in idle, it is normal you see it goes green when you power on the robots and then goes red after five minutes. Here is the checklist for diagnosing the issue:

- On Data Collector:
 - Are you able to login to Data Collector web interface with the assigned IP address? If not:
If the notebook PC has multiple network adaptors, make sure the active one has route to Data Collector. If there's a route but you are unable to access Data Collector web UI, you may have incorrect network configuration and need to redeploy the OVA.
 - Has ZXL site info file been installed?
 - If using DHCP, make sure the DHCP server is configured to assign Data Collector a fixed IP with correct subnet mask, gateway and DNS.
- Does robot controller have correct network settings?
 - **Host Comm** Network adaptor number, IP, subnet mask, router IP correct?
 - On ZDT settings, ZDT Data Collector IP address correct?
 - If Data Collector and robot controller are on different subnet, can the communication channel be established?
 - Any device in between blocks the traffic? Check R30iB, R30iA port requirements.

7.3.2 What if the Last Message Sent to Primary was Successful is Red?

Data Collector is unable to send the last message to Primary if this is red. Common issues are:

- Data Collector VM has no network communication permission.
- DNS failure.
- Incorrect network settings.
- HTTPS communication path to `zdtcdprimary.fanuczdt.com` or `fanuc01.cisco.com` can't be established.
- **Satellite-To-Primary** certificate has expired.
- If using HTTPS proxy, this could be due to incorrect HTTPS proxy setting on Data Collector or on proxy server.

7.4 Administrative Tasks

These are issues you might encounter when performing administrative tasks.

7.4.1 Why doesn't my company login work?

A role for your company login ID must be entered on the **User Management** page before you can log in with a company ID. This assumes that LDAP has been properly configured and tested during initial setup.

7.4.2 One or more of the ZDT processes on the ZDT Data Collector's Home page is not running

This is a diagnostic screen and you are probably looking at it because something seems to be wrong with the data collector. There is a log file associated with each of the zdt processes shown on the home screen. Go to the **Log Files** page. Find the associated process name in the left half of the screen. Focus on any log file without a .log extension since they are the most recent file for the specific data collector process. Information at the end of the file may provide enough information for you to fix the problem. If not, contact FANUC.

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R

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- Robot Controller File Backup Configuration [71](#)

REVISION RECORD

Edition	Date	Contents
B	Apr., 2020	• Robot Controller File Backup Feature added. • EOAT Gripper Timing added. • Vision updates.
A	Nov., 2019	• First release of combined manual.

